

Functional Constraints

Flame competition constraint within pocket regions.

Penalty coefficient 0.0100

Maximum allowable heat flux density ratio (max/min) 1000.00

The linear burning rate of the first conditional fuel is greater than that of the second.

Penalty coefficient 0.0100

Constraint on maximum kinetic flame heat flux density.

Penalty coefficient 0.0100

Maximum kinetic flame heat flux density in inter-pocket medium 1.00e+09

Maximum kinetic flame heat flux density in skeleton structure 1.00e+08

Maximum kinetic flame heat flux density in pocket region outside skeleton 1.00e+08

Constraint on skeleton thickness to pore diameter ratio.

Penalty coefficient 0.0100

Threshold for skeleton thickness to pore diameter ratio 3.00

Constraint on skeleton thickness to large oxidizer particle diameter ratio.

Penalty coefficient 0.0100

Threshold for skeleton thickness to large oxidizer particle diameter ratio 1.00

Parametric Constraints

Constraint on allowable surface temperature range

Minimum surface temperature 600 K

Maximum surface temperature 900 K

Differential Evolution Algorithm

Population size: 468

Mutation force: 0.5

Crossover probability: 0.9

Number of threads: 26

Termination strategy: OrTerminationStrategy

Group Combustion Solver Parameters

Shared Parameters (All Compositions)

Pre-exponential coefficient for gas phase reactions in inter-pocket medium 158482065196.19614

Parameter bounds [1.000E+005; 1.000E+013]

Activation energy for gas phase reactions in inter-pocket medium 240331.6453536874

Parameter bounds [5.000E+004; 2.500E+005]

Pre-exponential factor for gas-phase reactions in pocket region outside skeleton 1401988.7162855444

Parameter bounds [1.000E+005; 1.000E+013]

Activation energy for gas-phase reactions in pocket region outside skeleton 50000.00021722986

Parameter bounds [5.000E+004; 2.500E+005]

Reaction order for gas-phase chemistry in inter-pocket medium 2.4509968252502694

Parameter bounds [0.000E+000; 2.500E+000]

Reaction order for gas-phase chemistry outside skeleton structure 0.483855427239117

Parameter bounds [0.000E+000; 2.500E+000]

Enthalpy difference between vaporization and binder decomposition 563542.8241395919

Parameter bounds [-1.000E+007; 1.000E+007]

Diffusion flame height correlation coefficient 2.3333

Parameter bounds [1.000E-003; 1.000E+001]

Power law exponent for burning rate dependency on A (skeleton thickness) 0.9594

Parameter bounds [0.000E+000; 3.000E+000]

Power law exponent for burning rate dependency on B (pore diameter) 0.9591

Parameter bounds [0.000E+000; 3.000E+000]

Radiation temperature correlation coefficient (average layer temperature) 0.0000

Parameter bounds [0.000E+000; 1.000E+000]

Diffusion-flame standoff pressure-factor C_{rxn} 0.00186521

Parameter bounds [0.000E+000; 5.000E+001]

Diffusion-flame standoff size-exponent m 0.107807

Parameter bounds [0.000E+000; 2.000E+000]

Bimodal-packing heat-feedback K_{pack} 8.06364

Parameter bounds [0.000E+000; 1.000E+001]
Diffusion-flame standoff pressure-exponent n_p 4
Parameter bounds [2.000E+000; 4.000E+000]
WSB condensed-phase reaction K_{wsb} 0.220748
Parameter bounds [0.000E+000; 5.000E+000]
AP premixed-flame conductance K_{AP} 1487.58
Parameter bounds [0.000E+000; 5.000E+003]
AP self-deflagration pressure-exponent n_{AP} 1.75391
Parameter bounds [7.700E-001; 2.500E+000]

Bas_2 (includes Bas_3, Bas_4) - Condensed Phase Parameters

Pre-exponential coefficient for condensed phase reactions 9999839501713.129
Parameter bounds [1.000E+000; 1.000E+013]
Activation energy for condensed phase reactions 184330.9467568574
Parameter bounds [5.000E+004; 3.000E+005]
Pre-exponential factor for gas-phase reactions within skeleton structure 510820.5252288952
Parameter bounds [1.000E+005; 1.000E+013]
Activation energy for gas-phase reactions within skeleton structure 81897.7697913219
Parameter bounds [5.000E+004; 2.500E+005]
Reaction order for gas-phase chemistry in skeleton structure 0.49652649262917214
Parameter bounds [0.000E+000; 2.500E+000]
Coefficient A for metal combustion heat flux density in skeleton structure, 4.5899794457636127E-07
Parameter bounds [1.000E-010; 1.000E-003]
Coefficient B for metal combustion heat flux density in skeleton structure, 1.0276271017221015E-12
Parameter bounds [1.000E-012; 1.000E-005]

Bas_1 - Condensed Phase Parameters

Pre-exponential coefficient for condensed phase reactions 9999529360574.121
Parameter bounds [1.000E+000; 1.000E+013]
Activation energy for condensed phase reactions 141943.53120649775
Parameter bounds [5.000E+004; 3.000E+005]
Pre-exponential factor for gas-phase reactions within skeleton structure 38522787.944583766
Parameter bounds [1.000E+005; 1.000E+013]
Activation energy for gas-phase reactions within skeleton structure 50002.244994733956
Parameter bounds [5.000E+004; 2.500E+005]
Reaction order for gas-phase chemistry in skeleton structure 1.4184435060641614
Parameter bounds [0.000E+000; 2.500E+000]
Coefficient A for metal combustion heat flux density in skeleton structure, 1.2414107976355286E-06
Parameter bounds [1.000E-010; 1.000E-003]
Coefficient B for metal combustion heat flux density in skeleton structure, 2.1746821063532531E-07
Parameter bounds [1.000E-012; 1.000E-005]

Bas_0 - Condensed Phase Parameters

Pre-exponential coefficient for condensed phase reactions 9999313878258.373
Parameter bounds [1.000E+000; 1.000E+013]
Activation energy for condensed phase reactions 167548.19000299074
Parameter bounds [5.000E+004; 3.000E+005]
Pre-exponential factor for gas-phase reactions within skeleton structure 133330722.29131109
Parameter bounds [1.000E+005; 1.000E+013]
Activation energy for gas-phase reactions within skeleton structure 50002.70760475377
Parameter bounds [5.000E+004; 2.500E+005]
Reaction order for gas-phase chemistry in skeleton structure 0.6033904681159155
Parameter bounds [0.000E+000; 2.500E+000]
Coefficient A for metal combustion heat flux density in skeleton structure, 6.861557440416799E-07
Parameter bounds [1.000E-010; 1.000E-003]
Coefficient B for metal combustion heat flux density in skeleton structure, 2.2899445551313423E-07
Parameter bounds [1.000E-012; 1.000E-005]

Optimization Results

Objective function value 0.0117

Total penalty from constraint violations 0.0000

Propellant Composition "Bas_2"

At pressure 1 MPa

Experimental burning rate 16.94 mm/s

Calculated burning rate 15.34 mm/s

First Conditional Fuel

Surface temperature 746.63 K

Linear burning rate 11.45 mm/s

Mass decomposition rate $18.99 \text{ kg}\cdot\text{s}^{-1}\cdot\text{m}^{-2}$

Kinetic flame heat flux density to surface $2.35\text{e}+07 \text{ W/m}^2$

Kinetic flame height 12.61 μm

Average kinetic flame density 1.78 kg/m^3

Average kinetic flame temperature 2,226.82 K

Binder sublimation heat flux density $2.35\text{e}+07 \text{ W/m}^2$

Surface heat balance error 0.0021 W/m^2

Secondary Conditional Fuel

Surface temperature 737.47 K

Linear burning rate 8.18 mm/s

Mass decomposition rate $13.58 \text{ kg}\cdot\text{s}^{-1}\cdot\text{m}^{-2}$

Region Outside Skeleton Structure

Kinetic flame heat flux density to surface $548,648.45 \text{ W/m}^2$

Kinetic flame height 296.64 μm

Average kinetic flame density 2.56 kg/m^3

Average kinetic flame temperature 1,551.23 K

Skeleton Structure Region

Kinetic flame heat flux density to surface $1\text{e}+07 \text{ W/m}^2$

Kinetic flame height 8.51 μm

Average kinetic flame density 3.41 kg/m^3

Average kinetic flame temperature 1,164.23 K

Heat flux density due to metal combustion, $3.59\text{e}+06 \text{ W/m}^2$

Effective thermal conductivity $0.44 \text{ W/(m}\cdot\text{K)}$

Conductive thermal conductivity $0.4 \text{ W/(m}\cdot\text{K)}$

Radiative thermal conductivity $0.037 \text{ W/(m}\cdot\text{K)}$

Conductive thermal conductivity balance error $-4.092294297874943\text{E}-08$

Skeleton layer thickness $6.9\text{e}-05 \text{ m}$

Pore diameter $2.3\text{e}-05 \text{ m}$

Diffusion flame heat flux density $43,562.93 \text{ W/m}^2$

Diffusion flame height 6,580.21 μm

Heat flux density to surface within skeletal structure $4.89\text{e}+06 \text{ W/m}^2$

Heat flux density to surface outside skeletal structure $351,720.7 \text{ W/m}^2$

Total heat flux density to surface $1.5\text{e}+07 \text{ W/m}^2$

Binder sublimation heat flux density $1.5\text{e}+07 \text{ W/m}^2$

Surface heat balance error -0.0029 W/m^2

At pressure 4.06 MPa

Experimental burning rate 35.09 mm/s

Calculated burning rate 36.16 mm/s

First Conditional Fuel

Surface temperature 797.98 K

Linear burning rate 65 mm/s

Mass decomposition rate $107.83 \text{ kg}\cdot\text{s}^{-1}\cdot\text{m}^{-2}$

Kinetic flame heat flux density to surface $1.42\text{e}+08 \text{ W/m}^2$

Kinetic flame height 2.05 μm

Average kinetic flame density 7.15 kg/m^3

Average kinetic flame temperature 2,252.49 K

Binder sublimation heat flux density $1.42\text{e}+08 \text{ W/m}^2$
Surface heat balance error -0.024 W/m^2

Secondary Conditional Fuel

Surface temperature 756.65 K
Linear burning rate 16.36 mm/s
Mass decomposition rate $27.14 \text{ kg}\cdot\text{s}^{-1}\cdot\text{m}^{-2}$

Region Outside Skeleton Structure

Kinetic flame heat flux density to surface $545,194.45 \text{ W/m}^2$
Kinetic flame height $295.01 \text{ }\mu\text{m}$
Average kinetic flame density 10.31 kg/m^3
Average kinetic flame temperature $1,560.82 \text{ K}$

Skeleton Structure Region

Kinetic flame heat flux density to surface $1.19\text{e}+07 \text{ W/m}^2$
Kinetic flame height $7.04 \text{ }\mu\text{m}$
Average kinetic flame density 13.71 kg/m^3
Average kinetic flame temperature $1,173.82 \text{ K}$
Heat flux density due to metal combustion, $6.47\text{e}+06 \text{ W/m}^2$
Effective thermal conductivity $0.42 \text{ W/(m}\cdot\text{K)}$
Conductive thermal conductivity $0.4 \text{ W/(m}\cdot\text{K)}$
Radiative thermal conductivity $0.019 \text{ W/(m}\cdot\text{K)}$
Conductive thermal conductivity balance error $-4.092294297874943\text{E-}08$
Skeleton layer thickness $3.55\text{e-}05 \text{ m}$
Pore diameter $1.18\text{e-}05 \text{ m}$
Diffusion flame heat flux density $111,360.09 \text{ W/m}^2$
Diffusion flame height $2,556.89 \text{ }\mu\text{m}$
Heat flux density to surface within skeletal structure $6.05\text{e}+06 \text{ W/m}^2$
Heat flux density to surface outside skeletal structure $365,091.49 \text{ W/m}^2$
Total heat flux density to surface $1.93\text{e}+07 \text{ W/m}^2$
Binder sublimation heat flux density $1.93\text{e}+07 \text{ W/m}^2$
Surface heat balance error 0.0027 W/m^2

At pressure 6.5 MPa

Experimental burning rate 44.85 mm/s
Calculated burning rate 43.58 mm/s

First Conditional Fuel

Surface temperature 816.93 K
Linear burning rate 116.77 mm/s
Mass decomposition rate $193.72 \text{ kg}\cdot\text{s}^{-1}\cdot\text{m}^{-2}$
Kinetic flame heat flux density to surface $2.6\text{e}+08 \text{ W/m}^2$
Kinetic flame height $1.11 \text{ }\mu\text{m}$
Average kinetic flame density 11.41 kg/m^3
Average kinetic flame temperature $2,261.97 \text{ K}$
Binder sublimation heat flux density $2.6\text{e}+08 \text{ W/m}^2$
Surface heat balance error -0.029 W/m^2

Secondary Conditional Fuel

Surface temperature 760.98 K
Linear burning rate 19.04 mm/s
Mass decomposition rate $31.59 \text{ kg}\cdot\text{s}^{-1}\cdot\text{m}^{-2}$

Region Outside Skeleton Structure

Kinetic flame heat flux density to surface $589,698.66 \text{ W/m}^2$
Kinetic flame height $272.01 \text{ }\mu\text{m}$
Average kinetic flame density 16.51 kg/m^3
Average kinetic flame temperature $1,562.99 \text{ K}$

Skeleton Structure Region

Kinetic flame heat flux density to surface $1.36\text{e}+07 \text{ W/m}^2$
Kinetic flame height $6.11 \text{ }\mu\text{m}$
Average kinetic flame density 21.94 kg/m^3
Average kinetic flame temperature $1,175.99 \text{ K}$
Heat flux density due to metal combustion, $7.38\text{e}+06 \text{ W/m}^2$

Effective thermal conductivity 0.42 W/(m·K)
 Conductive thermal conductivity 0.4 W/(m·K)
 Radiative thermal conductivity 0.016 W/(m·K)
 Conductive thermal conductivity balance error -4.092294297874943E-08
 Skeleton layer thickness 3.07e-05 m
 Pore diameter 1.02e-05 m
 Diffusion flame heat flux density 96,800.88 W/m²
 Diffusion flame height 2,936.97 μm
 Heat flux density to surface within skeletal structure 5.69e+06 W/m²
 Heat flux density to surface outside skeletal structure 429,489.48 W/m²
 Total heat flux density to surface 1.99e+07 W/m²
 Binder sublimation heat flux density 1.99e+07 W/m²
 Surface heat balance error -0.0015 W/m²

Propellant Composition "Bas_3"

At pressure 1 MPa

Experimental burning rate 15.24 mm/s
 Calculated burning rate 15.28 mm/s

First Conditional Fuel

Surface temperature 746.63 K
 Linear burning rate 11.45 mm/s
 Mass decomposition rate 18.99 kg·s⁻¹·m⁻²
 Kinetic flame heat flux density to surface 2.35e+07 W/m²
 Kinetic flame height 12.61 μm
 Average kinetic flame density 1.78 kg/m³
 Average kinetic flame temperature 2,226.82 K
 Binder sublimation heat flux density 2.35e+07 W/m²
 Surface heat balance error 0.0021 W/m²

Secondary Conditional Fuel

Surface temperature 746.63 K
 Linear burning rate 11.45 mm/s
 Mass decomposition rate 18.99 kg·s⁻¹·m⁻²

Region Outside Skeleton Structure

Kinetic flame heat flux density to surface 393,988.84 W/m²
 Kinetic flame height 410.76 μm
 Average kinetic flame density 2.55 kg/m³
 Average kinetic flame temperature 1,555.82 K

Skeleton Structure Region

Kinetic flame heat flux density to surface 6.22e+07 W/m²
 Kinetic flame height 3.01 μm
 Average kinetic flame density 2.36 kg/m³
 Average kinetic flame temperature 1,684.82 K
 Heat flux density due to metal combustion, 4.28e+06 W/m²
 Effective thermal conductivity 0.39 W/(m·K)
 Conductive thermal conductivity 0.22 W/(m·K)
 Radiative thermal conductivity 0.17 W/(m·K)
 Conductive thermal conductivity balance error -5.5152085010057306E-08
 Skeleton layer thickness 5e-05 m
 Pore diameter 1.67e-05 m
 Diffusion flame heat flux density 80,647.98 W/m²
 Diffusion flame height 3,629.81 μm
 Heat flux density to surface within skeletal structure 2.08e+07 W/m²
 Heat flux density to surface outside skeletal structure 270,705.6 W/m²
 Total heat flux density to surface 2.12e+07 W/m²
 Binder sublimation heat flux density 2.12e+07 W/m²
 Surface heat balance error -0.0052 W/m²

At pressure 4.06 MPa

Experimental burning rate 37.34 mm/s
Calculated burning rate 38.03 mm/s

First Conditional Fuel

Surface temperature 797.98 K
Linear burning rate 65 mm/s
Mass decomposition rate $107.83 \text{ kg}\cdot\text{s}^{-1}\cdot\text{m}^{-2}$
Kinetic flame heat flux density to surface $1.42\text{e}+08 \text{ W/m}^2$
Kinetic flame height $2.05 \mu\text{m}$
Average kinetic flame density 7.15 kg/m^3
Average kinetic flame temperature 2,252.49 K
Binder sublimation heat flux density $1.42\text{e}+08 \text{ W/m}^2$
Surface heat balance error -0.024 W/m^2

Secondary Conditional Fuel

Surface temperature 762.41 K
Linear burning rate 20.01 mm/s
Mass decomposition rate $33.19 \text{ kg}\cdot\text{s}^{-1}\cdot\text{m}^{-2}$

Region Outside Skeleton Structure

Kinetic flame heat flux density to surface $446,982.47 \text{ W/m}^2$
Kinetic flame height $358.54 \mu\text{m}$
Average kinetic flame density 10.29 kg/m^3
Average kinetic flame temperature 1,563.7 K

Skeleton Structure Region

Kinetic flame heat flux density to surface $8.33\text{e}+07 \text{ W/m}^2$
Kinetic flame height $2.23 \mu\text{m}$
Average kinetic flame density 9.51 kg/m^3
Average kinetic flame temperature 1,692.7 K
Heat flux density due to metal combustion, $5.85\text{e}+06 \text{ W/m}^2$
Effective thermal conductivity $0.32 \text{ W/(m}\cdot\text{K)}$
Conductive thermal conductivity $0.22 \text{ W/(m}\cdot\text{K)}$
Radiative thermal conductivity $0.097 \text{ W/(m}\cdot\text{K)}$
Conductive thermal conductivity balance error $-5.5152085010057306\text{E-}08$
Skeleton layer thickness $2.93\text{e-}05 \text{ m}$
Pore diameter $9.75\text{e-}06 \text{ m}$

Diffusion flame heat flux density $233,011.65 \text{ W/m}^2$
Diffusion flame height $1,249.55 \mu\text{m}$
Heat flux density to surface within skeletal structure $2.08\text{e}+07 \text{ W/m}^2$
Heat flux density to surface outside skeletal structure $342,755 \text{ W/m}^2$
Total heat flux density to surface $2.37\text{e}+07 \text{ W/m}^2$
Binder sublimation heat flux density $2.37\text{e}+07 \text{ W/m}^2$
Surface heat balance error 0.0048 W/m^2

At pressure 6.5 MPa

Experimental burning rate 50.5 mm/s
Calculated burning rate 45.44 mm/s

First Conditional Fuel

Surface temperature 816.93 K
Linear burning rate 116.77 mm/s
Mass decomposition rate $193.72 \text{ kg}\cdot\text{s}^{-1}\cdot\text{m}^{-2}$
Kinetic flame heat flux density to surface $2.6\text{e}+08 \text{ W/m}^2$
Kinetic flame height $1.11 \mu\text{m}$
Average kinetic flame density 11.41 kg/m^3
Average kinetic flame temperature 2,261.97 K
Binder sublimation heat flux density $2.6\text{e}+08 \text{ W/m}^2$
Surface heat balance error -0.029 W/m^2

Secondary Conditional Fuel

Surface temperature 765.4 K
Linear burning rate 22.18 mm/s
Mass decomposition rate $36.8 \text{ kg}\cdot\text{s}^{-1}\cdot\text{m}^{-2}$
Region Outside Skeleton Structure

Kinetic flame heat flux density to surface 507,204.62 W/m²
Kinetic flame height 315.38 μm
Average kinetic flame density 16.48 kg/m³
Average kinetic flame temperature 1,565.2 K

Skeleton Structure Region

Kinetic flame heat flux density to surface 1e+08 W/m²
Kinetic flame height 1.86 μm
Average kinetic flame density 15.23 kg/m³
Average kinetic flame temperature 1,694.2 K
Heat flux density due to metal combustion, 6.24e+06 W/m²
Effective thermal conductivity 0.31 W/(m·K)
Conductive thermal conductivity 0.22 W/(m·K)
Radiative thermal conductivity 0.087 W/(m·K)
Conductive thermal conductivity balance error -5.5152085010057306E-08
Skeleton layer thickness 2.65e-05 m
Pore diameter 8.83e-06 m

Diffusion flame heat flux density 212,680.89 W/m²
Diffusion flame height 1,367.59 μm
Heat flux density to surface within skeletal structure 1.61e+07 W/m²
Heat flux density to surface outside skeletal structure 430,450.85 W/m²
Total heat flux density to surface 2.32e+07 W/m²
Binder sublimation heat flux density 2.32e+07 W/m²
Surface heat balance error -0.0057 W/m²

Propellant Composition "Bas_4"

At pressure 1 MPa

Experimental burning rate 9.73 mm/s
Calculated burning rate 9.98 mm/s

First Conditional Fuel

Surface temperature 746.63 K
Linear burning rate 11.45 mm/s
Mass decomposition rate 18.99 kg·s⁻¹·m⁻²
Kinetic flame heat flux density to surface 2.35e+07 W/m²
Kinetic flame height 12.61 μm
Average kinetic flame density 1.78 kg/m³
Average kinetic flame temperature 2,226.82 K
Binder sublimation heat flux density 2.35e+07 W/m²
Surface heat balance error 0.0021 W/m²

Secondary Conditional Fuel

Surface temperature 714.56 K
Linear burning rate 3.41 mm/s
Mass decomposition rate 5.65 kg·s⁻¹·m⁻²

Region Outside Skeleton Structure

Kinetic flame heat flux density to surface 17,899.54 W/m²
Kinetic flame height 1,829.33 μm
Average kinetic flame density 4.52 kg/m³
Average kinetic flame temperature 878.28 K

Skeleton Structure Region

Kinetic flame heat flux density to surface 6.16e+06 W/m²
Kinetic flame height 8.54 μm
Average kinetic flame density 4.06 kg/m³
Average kinetic flame temperature 977.78 K
Heat flux density due to metal combustion, 5.28e+06 W/m²
Effective thermal conductivity 1.44 W/(m·K)
Conductive thermal conductivity 1.41 W/(m·K)
Radiative thermal conductivity 0.032 W/(m·K)
Conductive thermal conductivity balance error 9.240271880983641E-10
Skeleton layer thickness 1.6e-04 m

Pore diameter $5.33\text{e-}05$ m
Diffusion flame heat flux density $18,274.99$ W/m²
Diffusion flame height $15,679.58$ μm
Heat flux density to surface within skeletal structure $6.05\text{e+}06$ W/m²
Heat flux density to surface outside skeletal structure $8,434.84$ W/m²
Total heat flux density to surface $6.08\text{e+}06$ W/m²
Binder sublimation heat flux density $6.08\text{e+}06$ W/m²
Surface heat balance error $7.48\text{e-}04$ W/m²

At pressure 4.06 MPa

Experimental burning rate 25.92 mm/s
Calculated burning rate 26.14 mm/s

First Conditional Fuel

Surface temperature 797.98 K
Linear burning rate 65 mm/s
Mass decomposition rate 107.83 kg·s⁻¹·m⁻²
Kinetic flame heat flux density to surface $1.42\text{e+}08$ W/m²
Kinetic flame height 2.05 μm
Average kinetic flame density 7.15 kg/m³
Average kinetic flame temperature $2,252.49$ K
Binder sublimation heat flux density $1.42\text{e+}08$ W/m²
Surface heat balance error -0.024 W/m²

Secondary Conditional Fuel

Surface temperature 736.6 K
Linear burning rate 7.92 mm/s
Mass decomposition rate 13.15 kg·s⁻¹·m⁻²

Region Outside Skeleton Structure

Kinetic flame heat flux density to surface $15,295.34$ W/m²
Kinetic flame height $1,996.7$ μm
Average kinetic flame density 18.1 kg/m³
Average kinetic flame temperature 889.3 K

Skeleton Structure Region

Kinetic flame heat flux density to surface $6.29\text{e+}06$ W/m²
Kinetic flame height 8.02 μm
Average kinetic flame density 16.28 kg/m³
Average kinetic flame temperature 988.8 K
Heat flux density due to metal combustion, $1.13\text{e+}07$ W/m²
Effective thermal conductivity 1.42 W/(m·K)
Conductive thermal conductivity 1.41 W/(m·K)
Radiative thermal conductivity 0.014 W/(m·K)
Conductive thermal conductivity balance error $9.240271880983641\text{E-}10$
Skeleton layer thickness $7.11\text{e-}05$ m
Pore diameter $2.37\text{e-}05$ m
Diffusion flame heat flux density $44,231.26$ W/m²
Diffusion flame height $6,428.49$ μm
Heat flux density to surface within skeletal structure $9.22\text{e+}06$ W/m²
Heat flux density to surface outside skeletal structure $7,265.6$ W/m²
Total heat flux density to surface $9.28\text{e+}06$ W/m²
Binder sublimation heat flux density $9.28\text{e+}06$ W/m²
Surface heat balance error 0.0011 W/m²

At pressure 6.5 MPa

Experimental burning rate 36.06 mm/s
Calculated burning rate 35.42 mm/s

First Conditional Fuel

Surface temperature 816.93 K
Linear burning rate 116.77 mm/s
Mass decomposition rate 193.72 kg·s⁻¹·m⁻²
Kinetic flame heat flux density to surface $2.6\text{e+}08$ W/m²

Kinetic flame height 1.11 μm
Average kinetic flame density 11.41 kg/m^3
Average kinetic flame temperature 2,261.97 K
Binder sublimation heat flux density 2.6e+08 W/m^2
Surface heat balance error -0.029 W/m^2

Secondary Conditional Fuel

Surface temperature 744.24 K
Linear burning rate 10.49 mm/s
Mass decomposition rate 17.41 $\text{kg}\cdot\text{s}^{-1}\cdot\text{m}^{-2}$

Region Outside Skeleton Structure

Kinetic flame heat flux density to surface 14,535.18 W/m^2
Kinetic flame height 2,048.58 μm
Average kinetic flame density 28.89 kg/m^3
Average kinetic flame temperature 893.12 K

Skeleton Structure Region

Kinetic flame heat flux density to surface 6.35e+06 W/m^2
Kinetic flame height 7.82 μm
Average kinetic flame density 25.99 kg/m^3
Average kinetic flame temperature 992.62 K
Heat flux density due to metal combustion, 1.45e+07 W/m^2
Effective thermal conductivity 1.42 $\text{W}/(\text{m}\cdot\text{K})$
Conductive thermal conductivity 1.41 $\text{W}/(\text{m}\cdot\text{K})$
Radiative thermal conductivity 0.011 $\text{W}/(\text{m}\cdot\text{K})$
Conductive thermal conductivity balance error 9.240271880983641E-10
Skeleton layer thickness 5.43e-05 m
Pore diameter 1.81e-05 m
Diffusion flame heat flux density 33,812.31 W/m^2
Diffusion flame height 8,386.78 μm
Heat flux density to surface within skeletal structure 1.09e+07 W/m^2
Heat flux density to surface outside skeletal structure 6,951.27 W/m^2
Total heat flux density to surface 1.09e+07 W/m^2
Binder sublimation heat flux density 1.09e+07 W/m^2
Surface heat balance error -9.95e-04 W/m^2

Propellant Composition "Bas_1"

At pressure 1 MPa

Experimental burning rate 14.26 mm/s
Calculated burning rate 14.25 mm/s

First Conditional Fuel

Surface temperature 631.46 K
Linear burning rate 10.93 mm/s
Mass decomposition rate 18.14 $\text{kg}\cdot\text{s}^{-1}\cdot\text{m}^{-2}$
Kinetic flame heat flux density to surface 1.93e+07 W/m^2
Kinetic flame height 15.94 μm
Average kinetic flame density 1.83 kg/m^3
Average kinetic flame temperature 2,169.23 K
Binder sublimation heat flux density 1.93e+07 W/m^2
Surface heat balance error 0.0034 W/m^2

Secondary Conditional Fuel

Surface temperature 622.9 K
Linear burning rate 7.54 mm/s
Mass decomposition rate 12.51 $\text{kg}\cdot\text{s}^{-1}\cdot\text{m}^{-2}$

Region Outside Skeleton Structure

Kinetic flame heat flux density to surface 559,397.44 W/m^2
Kinetic flame height 311.42 μm
Average kinetic flame density 2.66 kg/m^3
Average kinetic flame temperature 1,493.95 K

Skeleton Structure Region

Kinetic flame heat flux density to surface $7.97\text{e}+06 \text{ W/m}^2$
Kinetic flame height $12.14 \text{ }\mu\text{m}$
Average kinetic flame density 3.59 kg/m^3
Average kinetic flame temperature $1,106.95 \text{ K}$
Heat flux density due to metal combustion, $2.21\text{e}+06 \text{ W/m}^2$
Effective thermal conductivity $0.44 \text{ W/(m}\cdot\text{K)}$
Conductive thermal conductivity $0.4 \text{ W/(m}\cdot\text{K)}$
Radiative thermal conductivity $0.038 \text{ W/(m}\cdot\text{K)}$
Conductive thermal conductivity balance error $-4.092294297874943\text{E-}08$
Skeleton layer thickness $1.35\text{e-}04 \text{ m}$
Pore diameter $2.36\text{e-}05 \text{ m}$
Diffusion flame heat flux density $50,188.91 \text{ W/m}^2$
Diffusion flame height $6,063.29 \text{ }\mu\text{m}$
Heat flux density to surface within skeletal structure $3.85\text{e}+06 \text{ W/m}^2$
Heat flux density to surface outside skeletal structure $347,914.83 \text{ W/m}^2$
Total heat flux density to surface $1.2\text{e}+07 \text{ W/m}^2$
Binder sublimation heat flux density $1.2\text{e}+07 \text{ W/m}^2$
Surface heat balance error -0.0014 W/m^2

At pressure 4.06 MPa

Experimental burning rate 38.01 mm/s
Calculated burning rate 38.01 mm/s

First Conditional Fuel

Surface temperature 674.7 K
Linear burning rate 61.83 mm/s
Mass decomposition rate $102.58 \text{ kg}\cdot\text{s}^{-1}\cdot\text{m}^{-2}$
Kinetic flame heat flux density to surface $1.16\text{e}+08 \text{ W/m}^2$
Kinetic flame height $2.62 \text{ }\mu\text{m}$
Average kinetic flame density 7.35 kg/m^3
Average kinetic flame temperature $2,190.85 \text{ K}$
Binder sublimation heat flux density $1.16\text{e}+08 \text{ W/m}^2$
Surface heat balance error -0.011 W/m^2

Secondary Conditional Fuel

Surface temperature 642.49 K
Linear burning rate 17.39 mm/s
Mass decomposition rate $28.85 \text{ kg}\cdot\text{s}^{-1}\cdot\text{m}^{-2}$

Region Outside Skeleton Structure

Kinetic flame heat flux density to surface $483,145.95 \text{ W/m}^2$
Kinetic flame height $356.52 \text{ }\mu\text{m}$
Average kinetic flame density 10.7 kg/m^3
Average kinetic flame temperature $1,503.75 \text{ K}$

Skeleton Structure Region

Kinetic flame heat flux density to surface $2.56\text{e}+07 \text{ W/m}^2$
Kinetic flame height $3.71 \text{ }\mu\text{m}$
Average kinetic flame density 14.41 kg/m^3
Average kinetic flame temperature $1,116.75 \text{ K}$
Heat flux density due to metal combustion, $4.57\text{e}+06 \text{ W/m}^2$
Effective thermal conductivity $0.42 \text{ W/(m}\cdot\text{K)}$
Conductive thermal conductivity $0.4 \text{ W/(m}\cdot\text{K)}$
Radiative thermal conductivity $0.017 \text{ W/(m}\cdot\text{K)}$
Conductive thermal conductivity balance error $-4.092294297874943\text{E-}08$
Skeleton layer thickness $6.06\text{e-}05 \text{ m}$
Pore diameter $1.06\text{e-}05 \text{ m}$
Diffusion flame heat flux density $111,232.12 \text{ W/m}^2$
Diffusion flame height $2,718.2 \text{ }\mu\text{m}$
Heat flux density to surface within skeletal structure $6.03\text{e}+06 \text{ W/m}^2$
Heat flux density to surface outside skeletal structure $386,373.64 \text{ W/m}^2$
Total heat flux density to surface $1.95\text{e}+07 \text{ W/m}^2$
Binder sublimation heat flux density $1.95\text{e}+07 \text{ W/m}^2$

Surface heat balance error 0.0052 W/m²

At pressure 6.5 MPa

Experimental burning rate 52.88 mm/s

Calculated burning rate 52.73 mm/s

First Conditional Fuel

Surface temperature 690.65 K

Linear burning rate 110.93 mm/s

Mass decomposition rate 184.03 kg·s⁻¹·m⁻²

Kinetic flame heat flux density to surface 2.12e+08 W/m²

Kinetic flame height 1.42 μm

Average kinetic flame density 11.73 kg/m³

Average kinetic flame temperature 2,198.83 K

Binder sublimation heat flux density 2.12e+08 W/m²

Surface heat balance error -0.019 W/m²

Secondary Conditional Fuel

Surface temperature 649.84 K

Linear burning rate 23.49 mm/s

Mass decomposition rate 38.97 kg·s⁻¹·m⁻²

Region Outside Skeleton Structure

Kinetic flame heat flux density to surface 451,428.14 W/m²

Kinetic flame height 379.94 μm

Average kinetic flame density 17.11 kg/m³

Average kinetic flame temperature 1,507.42 K

Skeleton Structure Region

Kinetic flame heat flux density to surface 3.71e+07 W/m²

Kinetic flame height 2.53 μm

Average kinetic flame density 23.03 kg/m³

Average kinetic flame temperature 1,120.42 K

Heat flux density due to metal combustion, 5.96e+06 W/m²

Effective thermal conductivity 0.42 W/(m·K)

Conductive thermal conductivity 0.4 W/(m·K)

Radiative thermal conductivity 0.013 W/(m·K)

Conductive thermal conductivity balance error -4.092294297874943E-08

Skeleton layer thickness 4.54e-05 m

Pore diameter 7.94e-06 m

Diffusion flame heat flux density 83,254.94 W/m²

Diffusion flame height 3,622.8 μm

Heat flux density to surface within skeletal structure 7.04e+06 W/m²

Heat flux density to surface outside skeletal structure 377,726 W/m²

Total heat flux density to surface 2.39e+07 W/m²

Binder sublimation heat flux density 2.39e+07 W/m²

Surface heat balance error -0.0061 W/m²

Propellant Composition "Bas_0"

At pressure 1 MPa

Experimental burning rate 5.97 mm/s

Calculated burning rate 5.97 mm/s

First Conditional Fuel

Surface temperature 822.36 K

Linear burning rate 11.81 mm/s

Mass decomposition rate 19.58 kg·s⁻¹·m⁻²

Kinetic flame heat flux density to surface 2.64e+07 W/m²

Kinetic flame height 10.91 μm

Average kinetic flame density 1.75 kg/m³

Average kinetic flame temperature 2,264.68 K

Binder sublimation heat flux density 2.64e+07 W/m²

Surface heat balance error 0.0048 W/m²

Secondary Conditional Fuel

Surface temperature 779.42 K

Linear burning rate 2.67 mm/s

Mass decomposition rate $4.43 \text{ kg} \cdot \text{s}^{-1} \cdot \text{m}^{-2}$

Region Outside Skeleton Structure

Kinetic flame heat flux density to surface $1.71\text{e}+06 \text{ W/m}^2$

Kinetic flame height 92.61 μm

Average kinetic flame density 2.52 kg/m^3

Average kinetic flame temperature 1,572.21 K

Skeleton Structure Region

Kinetic flame heat flux density to surface $4,188.23 \text{ W/m}^2$

Kinetic flame height 19,377.66 μm

Average kinetic flame density 3.35 kg/m^3

Average kinetic flame temperature 1,185.21 K

Heat flux density due to metal combustion, $1.56\text{e}+06 \text{ W/m}^2$

Effective thermal conductivity $0.4 \text{ W/(m} \cdot \text{K)}$

Conductive thermal conductivity $0.4 \text{ W/(m} \cdot \text{K)}$

Radiative thermal conductivity $4.86\text{e}-07 \text{ W/(m} \cdot \text{K)}$

Conductive thermal conductivity balance error $-4.092294297874943\text{E}-08$

Skeleton layer thickness $1.35\text{e}-04 \text{ m}$

Pore diameter $3.02\text{e}-10 \text{ m}$

Diffusion flame heat flux density $134,297.3 \text{ W/m}^2$

Diffusion flame height 2,149.4 μm

Heat flux density to surface within skeletal structure $403,535.52 \text{ W/m}^2$

Heat flux density to surface outside skeletal structure $1.27\text{e}+06 \text{ W/m}^2$

Total heat flux density to surface $5.12\text{e}+06 \text{ W/m}^2$

Binder sublimation heat flux density $5.12\text{e}+06 \text{ W/m}^2$

Surface heat balance error $-4.82\text{e}-04 \text{ W/m}^2$

At pressure 4.06 MPa

Experimental burning rate 13.63 mm/s

Calculated burning rate 13.63 mm/s

First Conditional Fuel

Surface temperature 879.07 K

Linear burning rate 67.19 mm/s

Mass decomposition rate $111.47 \text{ kg} \cdot \text{s}^{-1} \cdot \text{m}^{-2}$

Kinetic flame heat flux density to surface $1.6\text{e}+08 \text{ W/m}^2$

Kinetic flame height 1.77 μm

Average kinetic flame density 7.02 kg/m^3

Average kinetic flame temperature 2,293.03 K

Binder sublimation heat flux density $1.6\text{e}+08 \text{ W/m}^2$

Surface heat balance error -0.028 W/m^2

Secondary Conditional Fuel

Surface temperature 801.09 K

Linear burning rate 5.77 mm/s

Mass decomposition rate $9.57 \text{ kg} \cdot \text{s}^{-1} \cdot \text{m}^{-2}$

Region Outside Skeleton Structure

Kinetic flame heat flux density to surface $1.58\text{e}+06 \text{ W/m}^2$

Kinetic flame height 99.26 μm

Average kinetic flame density 10.17 kg/m^3

Average kinetic flame temperature 1,583.05 K

Skeleton Structure Region

Kinetic flame heat flux density to surface $4,061.5 \text{ W/m}^2$

Kinetic flame height 19,448.6 μm

Average kinetic flame density 13.46 kg/m^3

Average kinetic flame temperature 1,196.05 K

Heat flux density due to metal combustion, $3.12\text{e}+06 \text{ W/m}^2$

Effective thermal conductivity $0.4 \text{ W/(m} \cdot \text{K)}$

Conductive thermal conductivity $0.4 \text{ W/(m} \cdot \text{K)}$

Radiative thermal conductivity $2.32\text{e-}07 \text{ W/(m}\cdot\text{K)}$
Conductive thermal conductivity balance error $-4.092294297874943\text{E-}08$
Skeleton layer thickness $6.45\text{e-}05 \text{ m}$
Pore diameter $1.44\text{e-}10 \text{ m}$
Diffusion flame heat flux density $317,637.86 \text{ W/m}^2$
Diffusion flame height $901.94 \text{ }\mu\text{m}$
Heat flux density to surface within skeletal structure $586,145 \text{ W/m}^2$
Heat flux density to surface outside skeletal structure $1.28\text{e+}06 \text{ W/m}^2$
Total heat flux density to surface $6.96\text{e+}06 \text{ W/m}^2$
Binder sublimation heat flux density $6.96\text{e+}06 \text{ W/m}^2$
Surface heat balance error 0.0012 W/m^2

At pressure 6.5 MPa

Experimental burning rate 18 mm/s
Calculated burning rate 18.03 mm/s

First Conditional Fuel

Surface temperature 900 K
Linear burning rate 120.8 mm/s
Mass decomposition rate $200.41 \text{ kg}\cdot\text{s}^{-1}\cdot\text{m}^{-2}$
Kinetic flame heat flux density to surface $2.94\text{e+}08 \text{ W/m}^2$
Kinetic flame height $0.96 \text{ }\mu\text{m}$
Average kinetic flame density 11.2 kg/m^3
Average kinetic flame temperature $2,303.5 \text{ K}$
Binder sublimation heat flux density $2.94\text{e+}08 \text{ W/m}^2$
Surface heat balance error 0.04 W/m^2

Secondary Conditional Fuel

Surface temperature 808.98 K
Linear burning rate 7.56 mm/s
Mass decomposition rate $12.54 \text{ kg}\cdot\text{s}^{-1}\cdot\text{m}^{-2}$

Region Outside Skeleton Structure

Kinetic flame heat flux density to surface $1.52\text{e+}06 \text{ W/m}^2$
Kinetic flame height $102.63 \text{ }\mu\text{m}$
Average kinetic flame density 16.26 kg/m^3
Average kinetic flame temperature $1,586.99 \text{ K}$

Skeleton Structure Region

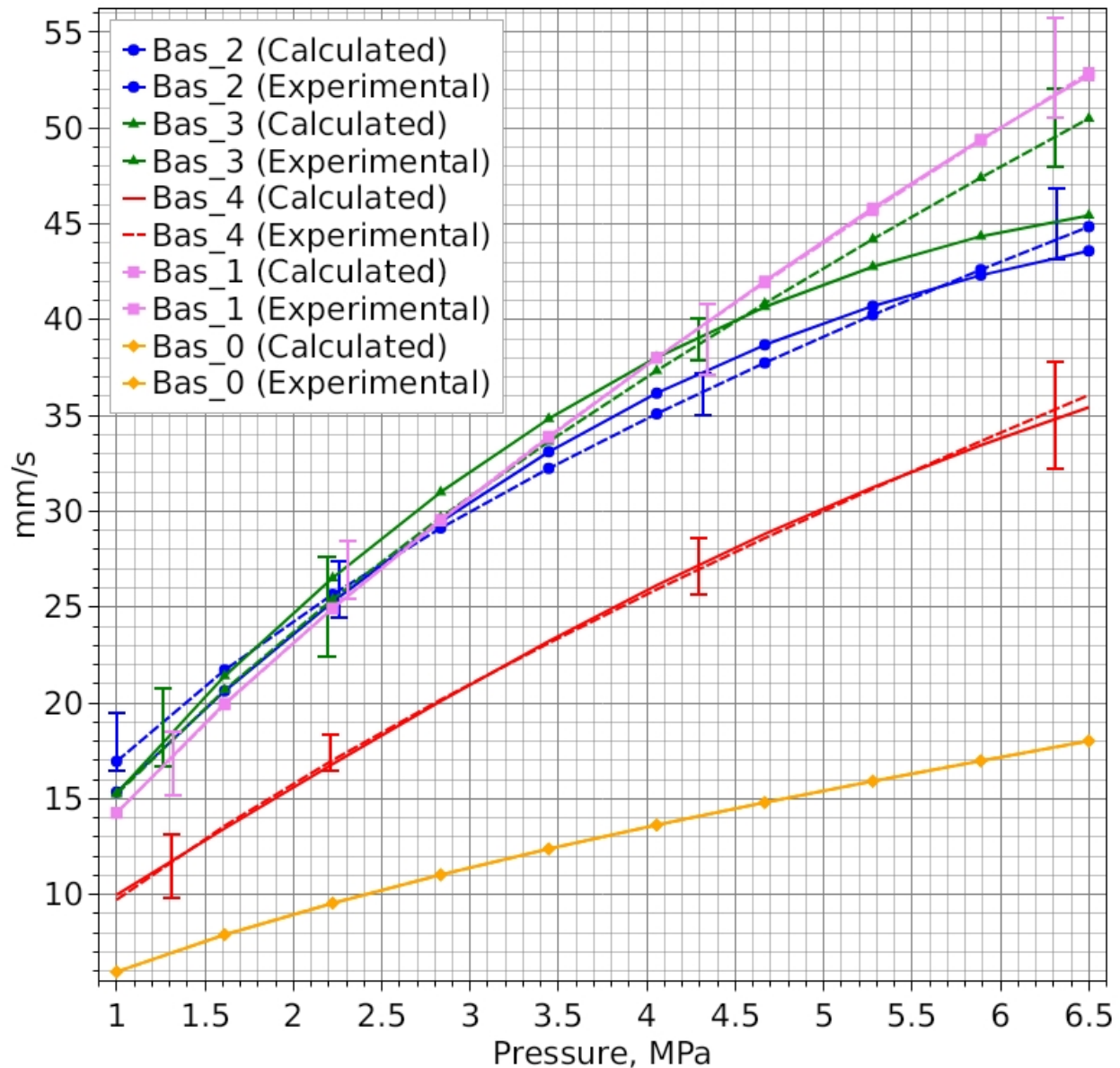
Kinetic flame heat flux density to surface $3,979.97 \text{ W/m}^2$
Kinetic flame height $19,648.79 \text{ }\mu\text{m}$
Average kinetic flame density 21.5 kg/m^3
Average kinetic flame temperature $1,199.99 \text{ K}$
Heat flux density due to metal combustion, $3.98\text{e+}06 \text{ W/m}^2$
Effective thermal conductivity $0.4 \text{ W/(m}\cdot\text{K)}$
Conductive thermal conductivity $0.4 \text{ W/(m}\cdot\text{K)}$
Radiative thermal conductivity $1.79\text{e-}07 \text{ W/(m}\cdot\text{K)}$
Conductive thermal conductivity balance error $-4.092294297874943\text{E-}08$
Skeleton layer thickness $4.98\text{e-}05 \text{ m}$
Pore diameter $1.11\text{e-}10 \text{ m}$
Diffusion flame heat flux density $245,038.03 \text{ W/m}^2$
Diffusion flame height $1,165.95 \text{ }\mu\text{m}$
Heat flux density to surface within skeletal structure $502,253.53 \text{ W/m}^2$
Heat flux density to surface outside skeletal structure $1.32\text{e+}06 \text{ W/m}^2$
Total heat flux density to surface $8\text{e+}06 \text{ W/m}^2$
Binder sublimation heat flux density $8\text{e+}06 \text{ W/m}^2$
Surface heat balance error $9.72\text{e-}05 \text{ W/m}^2$

1 MPa										
Parameter	Bas_2		Bas_3		Bas_4		Bas_1		Bas_0	
Experimental Burning Rate	16.94 mm/s		15.24 mm/s		9.73 mm/s		14.26 mm/s		5.97 mm/s	
Calculated Burning Rate	15.34 mm/s		15.28 mm/s		9.98 mm/s		14.25 mm/s		5.97 mm/s	
Difference	1.6 mm/s		-0.038 mm/s		-0.25 mm/s		0.012 mm/s		-0.0012 mm/s	
-	I	II	I	II	I	II	I	II	I	II
Linear Burning Rate	11.45 mm/s	8.18 mm/s	11.45 mm/s	11.45 mm/s	11.45 mm/s	3.41 mm/s	10.93 mm/s	7.54 mm/s	11.81 mm/s	2.67 mm/s
Surface Temperature	746.63 K	737.47 K	746.63 K	746.63 K	746.63 K	714.56 K	631.46 K	622.9 K	822.36 K	779.42 K
Kinetic Flame Heat Flux Density (Inter-Pocket)	2.35e+07 W/m ²		2.35e+07 W/m ²		2.35e+07 W/m ²		1.93e+07 W/m ²		2.64e+07 W/m ²	
Kinetic Flame Heat Flux Density (Skeleton)	1e+07 W/m ²		6.22e+07 W/m ²		6.16e+06 W/m ²		7.97e+06 W/m ²		4,188.23 W/m ²	
Kinetic Flame Height (Skeleton)	8.51 μm		3.01 μm		8.54 μm		12.14 μm		19,377.66 μm	
Kinetic Flame Heat Flux Density (Out-Skeleton)	548,648.45 W/m ²		393,988.84 W/m ²		17,899.54 W/m ²		559,397.44 W/m ²		1.71e+06 W/m ²	
Kinetic Flame Height (Out-Skeleton)	296.64 μm		410.76 μm		1,829.33 μm		311.42 μm		92.61 μm	
Diffusion Flame Heat Flux Density	43,562.93 W/m ²		80,647.98 W/m ²		18,274.99 W/m ²		50,188.91 W/m ²		134,297.3 W/m ²	
Diffusion Flame Height	6,580.21 μm		3,629.81 μm		15,679.58 μm		6,063.29 μm		2,149.4 μm	
Metal Combustion Heat Flux Density	3.59e+06 W/m ²		4.28e+06 W/m ²		5.28e+06 W/m ²		2.21e+06 W/m ²		1.56e+06 W/m ²	

4.06 MPa										
Parameter	Bas_2		Bas_3		Bas_4		Bas_1		Bas_0	
Experimental Burning Rate	35.09 mm/s		37.34 mm/s		25.92 mm/s		38.01 mm/s		13.63 mm/s	
Calculated Burning Rate	36.16 mm/s		38.03 mm/s		26.14 mm/s		38.01 mm/s		13.63 mm/s	
Difference	-1.07 mm/s		-0.69 mm/s		-0.21 mm/s		-4.83e-04 mm/s		-0.0034 mm/s	
-	I	II	I	II	I	II	I	II	I	II
Linear Burning Rate	65 mm/s	16.36 mm/s	65 mm/s	20.01 mm/s	65 mm/s	7.92 mm/s	61.83 mm/s	17.39 mm/s	67.19 mm/s	5.77 mm/s
Surface Temperature	797.98 K	756.65 K	797.98 K	762.41 K	797.98 K	736.6 K	674.7 K	642.49 K	879.07 K	801.09 K
Kinetic Flame Heat Flux Density (Inter-Pocket)	1.42e+08 W/m ²		1.42e+08 W/m ²		1.42e+08 W/m ²		1.16e+08 W/m ²		1.6e+08 W/m ²	
Kinetic Flame Heat Flux Density (Skeleton)	1.19e+07 W/m ²		8.33e+07 W/m ²		6.29e+06 W/m ²		2.56e+07 W/m ²		4,061.5 W/m ²	
Kinetic Flame Height (Skeleton)	7.04 μm		2.23 μm		8.02 μm		3.71 μm		19,448.6 μm	
Kinetic Flame Heat Flux Density (Out-Skeleton)	545,194.45 W/m ²		446,982.47 W/m ²		15,295.34 W/m ²		483,145.95 W/m ²		1.58e+06 W/m ²	
Kinetic Flame Height (Out-Skeleton)	295.01 μm		358.54 μm		1,996.7 μm		356.52 μm		99.26 μm	
Diffusion Flame Heat Flux Density	111,360.09 W/m ²		233,011.65 W/m ²		44,231.26 W/m ²		111,232.12 W/m ²		317,637.86 W/m ²	
Diffusion Flame Height	2,556.89 μm		1,249.55 μm		6,428.49 μm		2,718.2 μm		901.94 μm	
Metal Combustion Heat Flux Density	6.47e+06 W/m ²		5.85e+06 W/m ²		1.13e+07 W/m ²		4.57e+06 W/m ²		3.12e+06 W/m ²	

6.5 MPa										
Parameter	Bas_2		Bas_3		Bas_4		Bas_1		Bas_0	
Experimental Burning Rate	44.85 mm/s		50.5 mm/s		36.06 mm/s		52.88 mm/s		18 mm/s	
Calculated Burning Rate	43.58 mm/s		45.44 mm/s		35.42 mm/s		52.73 mm/s		18.03 mm/s	
Difference	1.26 mm/s		5.06 mm/s		0.64 mm/s		0.15 mm/s		-0.027 mm/s	
-	I	II	I	II	I	II	I	II	I	II
Linear Burning Rate	116.77 mm/s	19.04 mm/s	116.77 mm/s	22.18 mm/s	116.77 mm/s	10.49 mm/s	110.93 mm/s	23.49 mm/s	120.8 mm/s	7.56 mm/s
Surface Temperature	816.93 K	760.98 K	816.93 K	765.4 K	816.93 K	744.24 K	690.65 K	649.84 K	900 K	808.98 K
Kinetic Flame Heat Flux Density (Inter-Pocket)	2.6e+08 W/m²		2.6e+08 W/m²		2.6e+08 W/m²		2.12e+08 W/m²		2.94e+08 W/m²	
Kinetic Flame Heat Flux Density (Skeleton)	1.36e+07 W/m²		1e+08 W/m²		6.35e+06 W/m²		3.71e+07 W/m²		3,979.97 W/m²	
Kinetic Flame Height (Skeleton)	6.11 μm		1.86 μm		7.82 μm		2.53 μm		19,648.79 μm	
Kinetic Flame Heat Flux Density (Out-Skeleton)	589,698.66 W/m²		507,204.62 W/m²		14,535.18 W/m²		451,428.14 W/m²		1.52e+06 W/m²	
Kinetic Flame Height (Out-Skeleton)	272.01 μm		315.38 μm		2,048.58 μm		379.94 μm		102.63 μm	
Diffusion Flame Heat Flux Density	96,800.88 W/m²		212,680.89 W/m²		33,812.31 W/m²		83,254.94 W/m²		245,038.03 W/m²	
Diffusion Flame Height	2,936.97 μm		1,367.59 μm		8,386.78 μm		3,622.8 μm		1,165.95 μm	
Metal Combustion Heat Flux Density	7.38e+06 W/m²		6.24e+06 W/m²		1.45e+07 W/m²		5.96e+06 W/m²		3.98e+06 W/m²	

Burning Rates - Group Optimization (All Propellants)



Propellant Composition Parameters

Composition "Bas_2"

Density 1,659 kg/m³, specific heat capacity 1,500 J/kg.K, initial temperature 298 K, mass fraction of "pockets" in "binder-metal" composition 0.7.

Vieille's Power Law: $U = A \cdot p^v$, $A = 1.285341E-05$; $v = 0.52$.

Gas Phase Parameters in Inter-Pocket Medium

Thermal conductivity 0.1 W/(m·K)
 Average molar mass 0.033 kg/mol
 Specific heat capacity (constant volume) 635.6 J/kg.K
 Kinetic flame temperature 3,707 K

Gas Phase Parameters in Pocket Region

Thermal conductivity 0.1 W/(m·K)
 Average molar mass 0.033 kg/mol
 Specific heat capacity (constant volume) 635.6 J/kg.K
 Diffusion flame temperature 3,604 K

Gas Phase Parameters in Skeleton Structure

Thermal conductivity 0.1 W/(m·K)
 Average molar mass 0.033 kg/mol
 Specific heat capacity (constant volume) 635.6 J/kg.K
 Kinetic flame temperature 1,591 K

Gas Phase Parameters Outside Skeleton Structure

Thermal conductivity 0.1 W/(m·K)
Average molar mass 0.033 kg/mol
Specific heat capacity (constant volume) 635.6 J/kg.K
Kinetic flame temperature 2,365 K

Composition "Bas_3"

Density 1,659 kg/m³, specific heat capacity 1,500 J/kg.K, initial temperature 298 K, mass fraction of "pockets" in "binder-metal" composition 0.57.

Vieille's Power Law: $U = A \cdot p^v$, $A = 2.203002E-06$; $v = 0.64$.

Gas Phase Parameters in Inter-Pocket Medium

Thermal conductivity 0.1 W/(m·K)
Average molar mass 0.033 kg/mol
Specific heat capacity (constant volume) 635.6 J/kg.K
Kinetic flame temperature 3,707 K

Gas Phase Parameters in Pocket Region

Thermal conductivity 0.1 W/(m·K)
Average molar mass 0.033 kg/mol
Specific heat capacity (constant volume) 635.6 J/kg.K
Diffusion flame temperature 3,674 K

Gas Phase Parameters in Skeleton Structure

Thermal conductivity 0.1 W/(m·K)
Average molar mass 0.033 kg/mol
Specific heat capacity (constant volume) 635.6 J/kg.K
Kinetic flame temperature 2,623 K

Gas Phase Parameters Outside Skeleton Structure

Thermal conductivity 0.1 W/(m·K)
Average molar mass 0.033 kg/mol
Specific heat capacity (constant volume) 635.6 J/kg.K
Kinetic flame temperature 2,365 K

Composition "Bas_4"

Density 1,659 kg/m³, specific heat capacity 1,500 J/kg.K, initial temperature 298 K, mass fraction of "pockets" in "binder-metal" composition 0.56.

Vieille's Power Law: $U = A \cdot p^v$, $A = 6.137894E-07$; $v = 0.7$.

Gas Phase Parameters in Inter-Pocket Medium

Thermal conductivity 0.1 W/(m·K)
Average molar mass 0.033 kg/mol
Specific heat capacity (constant volume) 635.6 J/kg.K
Kinetic flame temperature 3,707 K

Gas Phase Parameters in Pocket Region

Thermal conductivity 0.1 W/(m·K)
Average molar mass 0.033 kg/mol
Specific heat capacity (constant volume) 635.6 J/kg.K
Diffusion flame temperature 3,580 K

Gas Phase Parameters in Skeleton Structure

Thermal conductivity 0.1 W/(m·K)
Average molar mass 0.033 kg/mol
Specific heat capacity (constant volume) 635.6 J/kg.K
Kinetic flame temperature 1,241 K

Gas Phase Parameters Outside Skeleton Structure

Thermal conductivity 0.1 W/(m·K)
Average molar mass 0.033 kg/mol
Specific heat capacity (constant volume) 635.6 J/kg.K
Kinetic flame temperature 1,042 K

Composition "Bas_1"

Density 1,659 kg/m³, specific heat capacity 1,500 J/kg.K, initial temperature 298 K, mass fraction of "pockets" in "binder-metal" composition 0.7.

Vieille's Power Law: $U = A \cdot p^v$, $A = 9.000448E-07$; $v = 0.7$.

Gas Phase Parameters in Inter-Pocket Medium

Thermal conductivity 0.1 W/(m·K)
Average molar mass 0.033 kg/mol

Specific heat capacity (constant volume) 635.6 J/kg.K

Kinetic flame temperature 3,707 K

Gas Phase Parameters in Pocket Region

Thermal conductivity 0.1 W/(m·K)

Average molar mass 0.033 kg/mol

Specific heat capacity (constant volume) 635.6 J/kg.K

Diffusion flame temperature 3,666 K

Gas Phase Parameters in Skeleton Structure

Thermal conductivity 0.1 W/(m·K)

Average molar mass 0.033 kg/mol

Specific heat capacity (constant volume) 635.6 J/kg.K

Kinetic flame temperature 1,591 K

Gas Phase Parameters Outside Skeleton Structure

Thermal conductivity 0.1 W/(m·K)

Average molar mass 0.033 kg/mol

Specific heat capacity (constant volume) 635.6 J/kg.K

Kinetic flame temperature 2,365 K

Composition "Bas_0"

Density 1,659 kg/m³, specific heat capacity 1,500 J/kg.K, initial temperature 298 K, mass fraction of "pockets" in "binder-metal" composition 0.7.

Vieille's Power Law: $U = A \cdot p^v$, $A = 1.720582E-06$; $v = 0.59$.

Gas Phase Parameters in Inter-Pocket Medium

Thermal conductivity 0.1 W/(m·K)

Average molar mass 0.033 kg/mol

Specific heat capacity (constant volume) 635.6 J/kg.K

Kinetic flame temperature 3,707 K

Gas Phase Parameters in Pocket Region

Thermal conductivity 0.1 W/(m·K)

Average molar mass 0.033 kg/mol

Specific heat capacity (constant volume) 635.6 J/kg.K

Diffusion flame temperature 3,666 K

Gas Phase Parameters in Skeleton Structure

Thermal conductivity 0.1 W/(m·K)

Average molar mass 0.033 kg/mol

Specific heat capacity (constant volume) 635.6 J/kg.K

Kinetic flame temperature 1,591 K

Gas Phase Parameters Outside Skeleton Structure

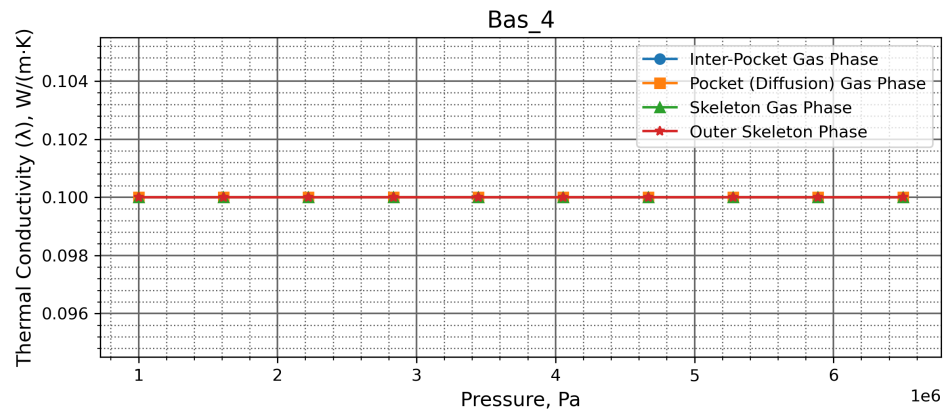
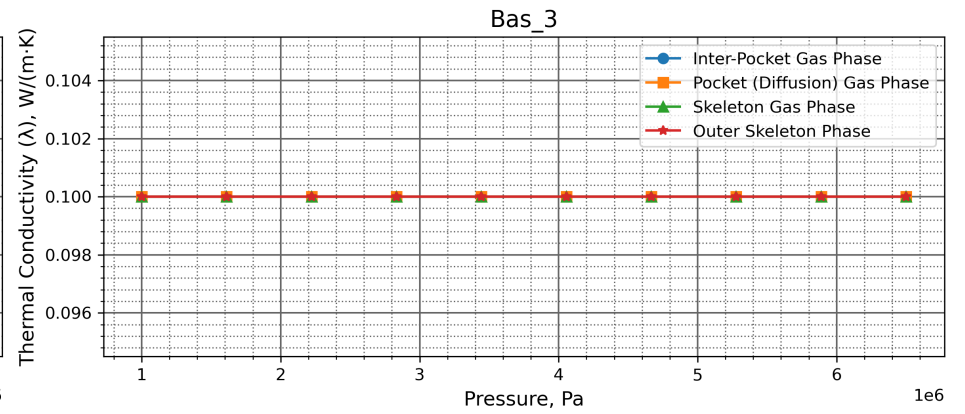
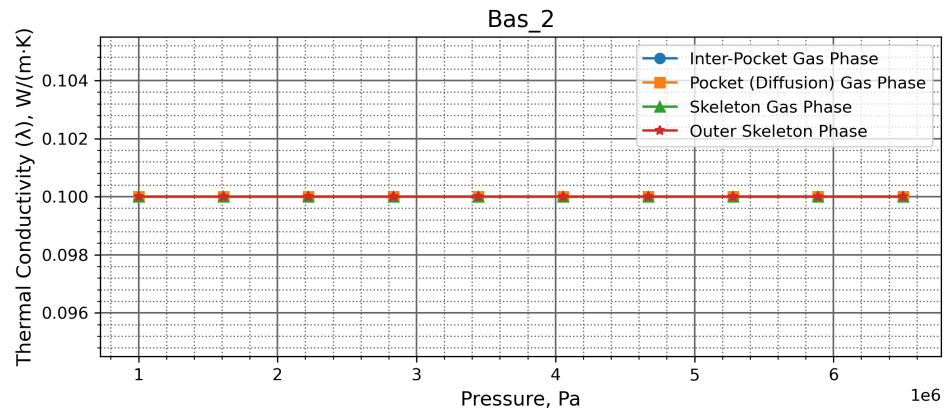
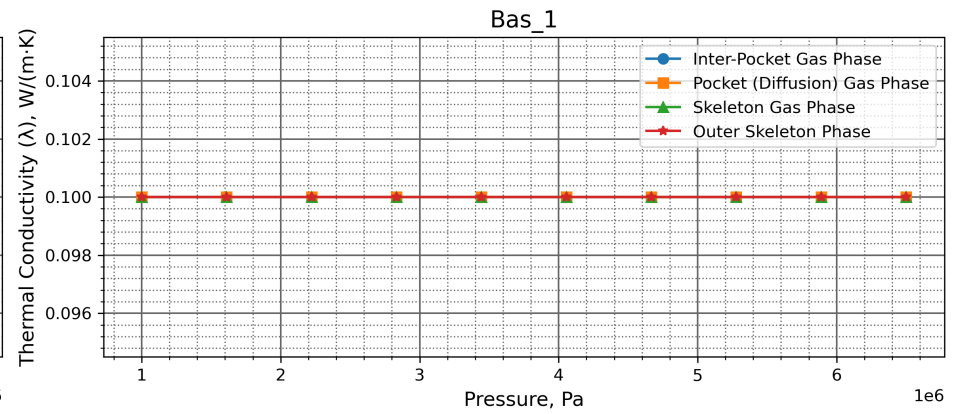
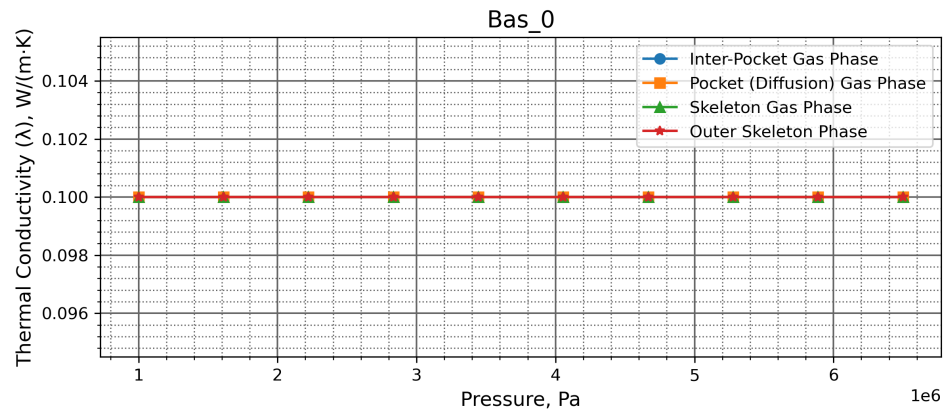
Thermal conductivity 0.1 W/(m·K)

Average molar mass 0.033 kg/mol

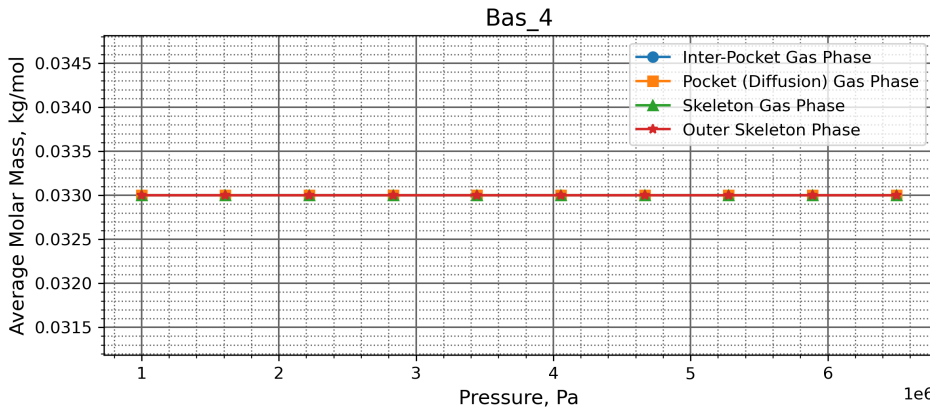
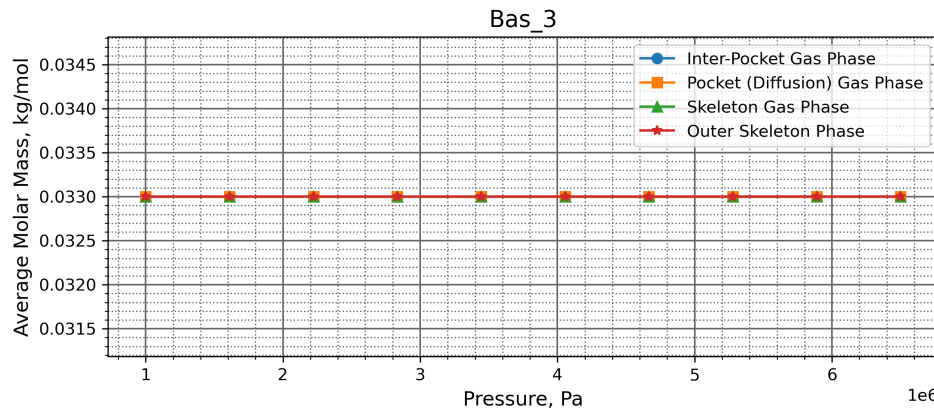
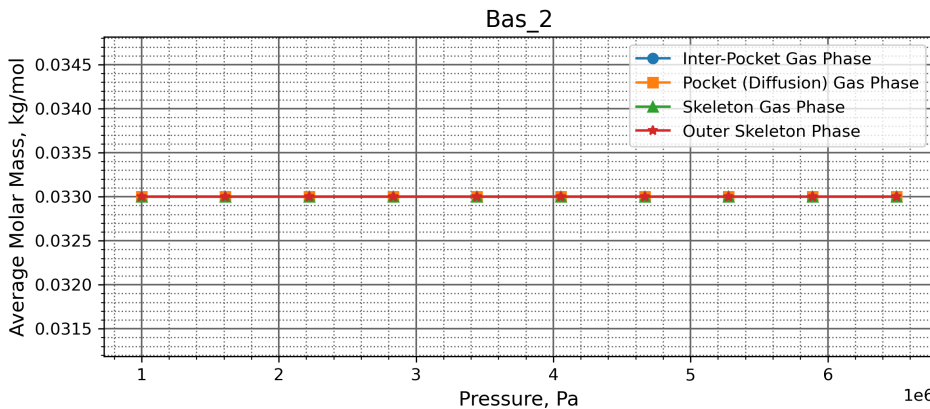
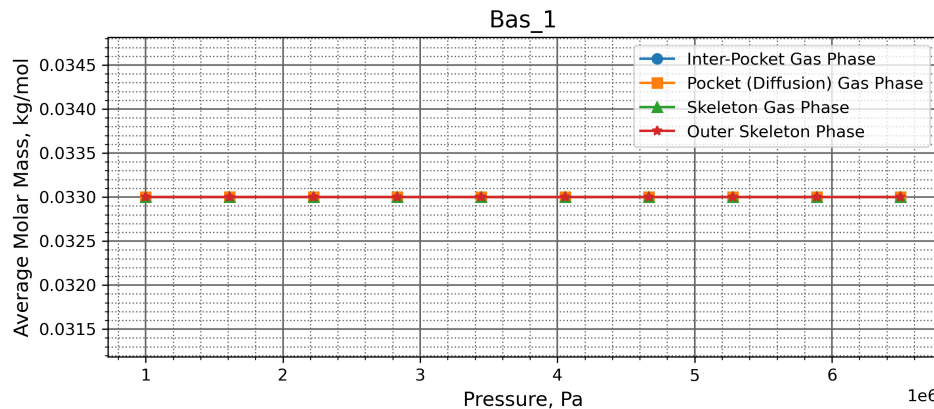
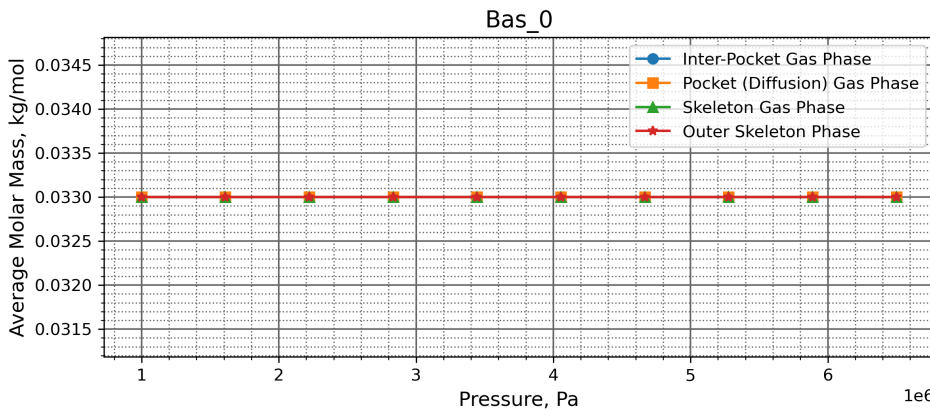
Specific heat capacity (constant volume) 635.6 J/kg.K

Kinetic flame temperature 2,365 K

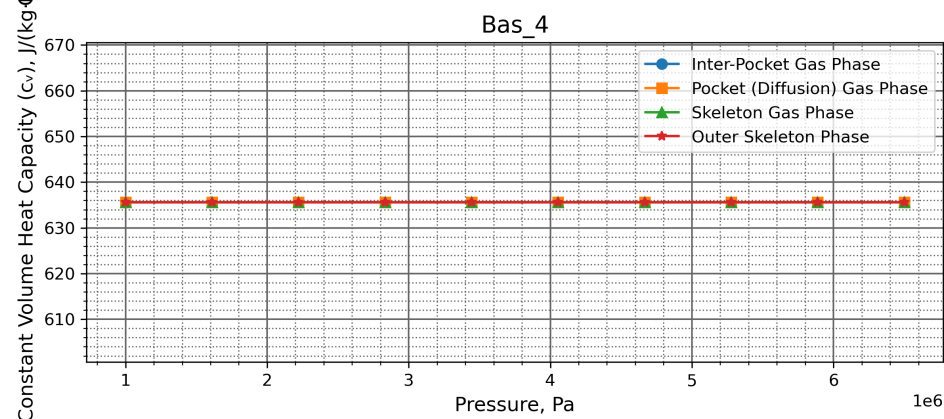
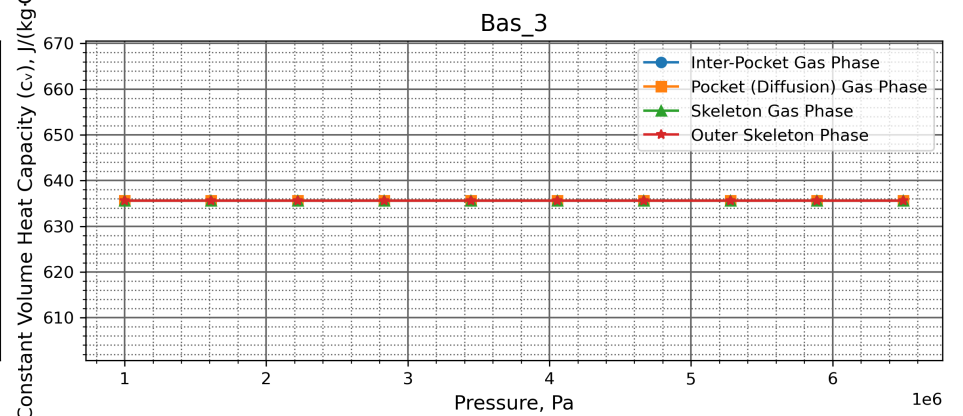
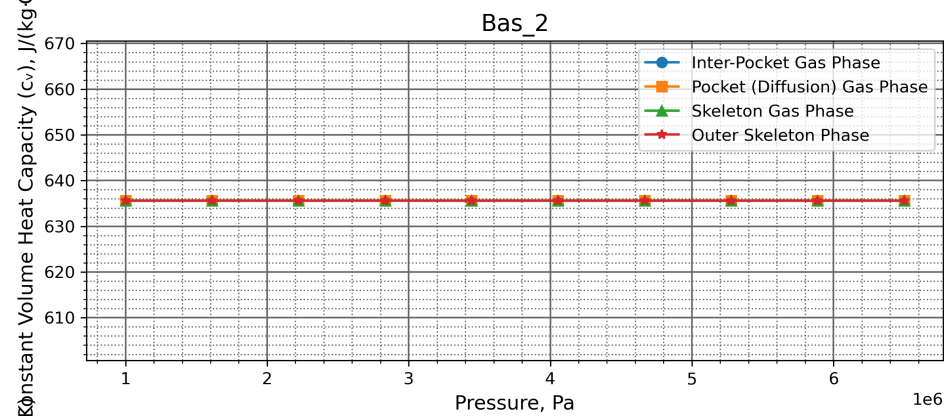
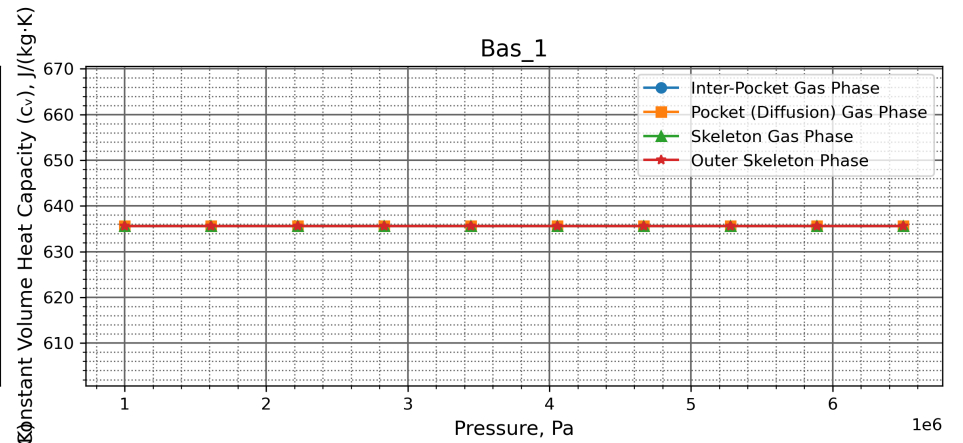
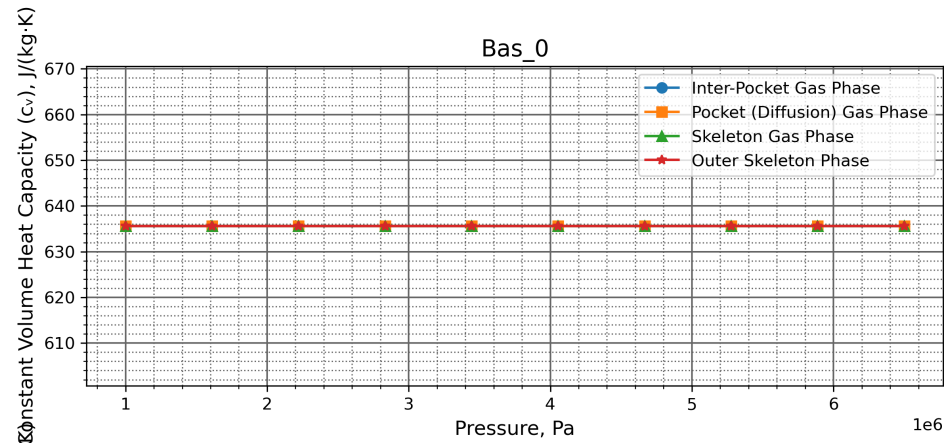
Thermal Conductivity (λ), W/(m·K)



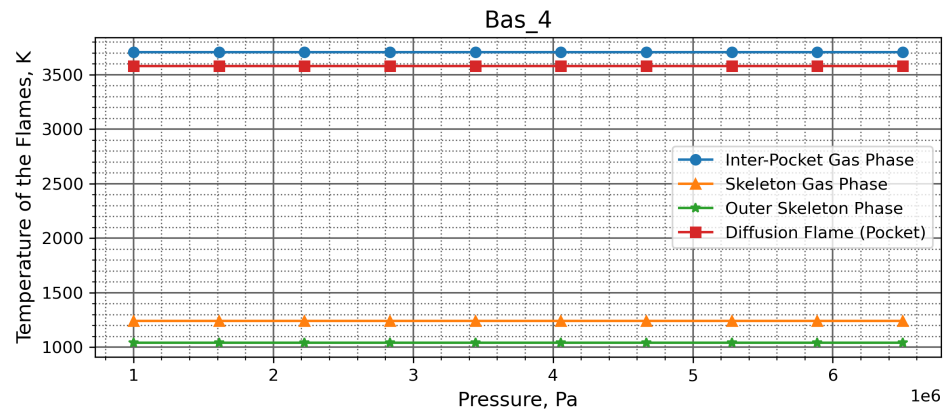
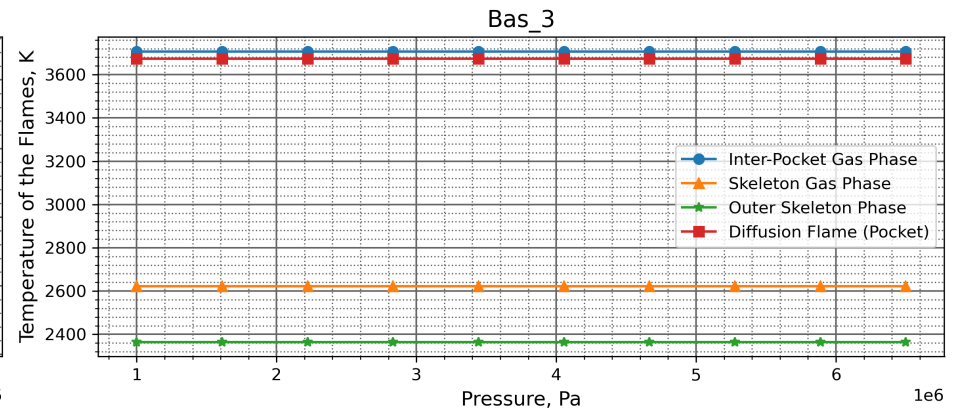
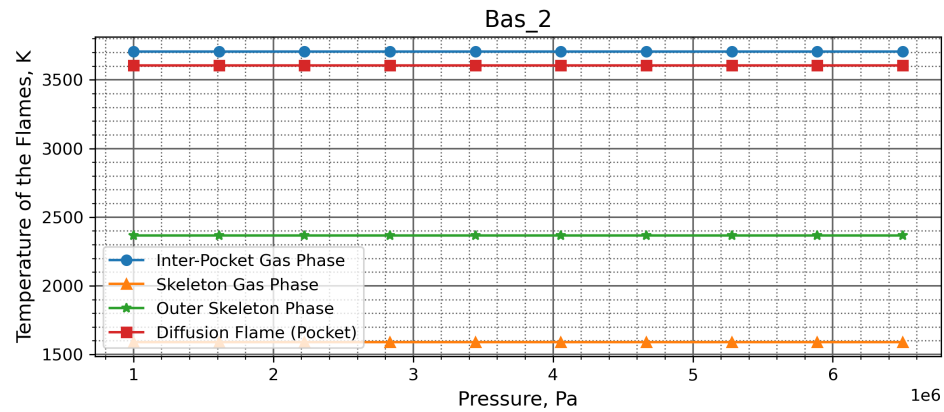
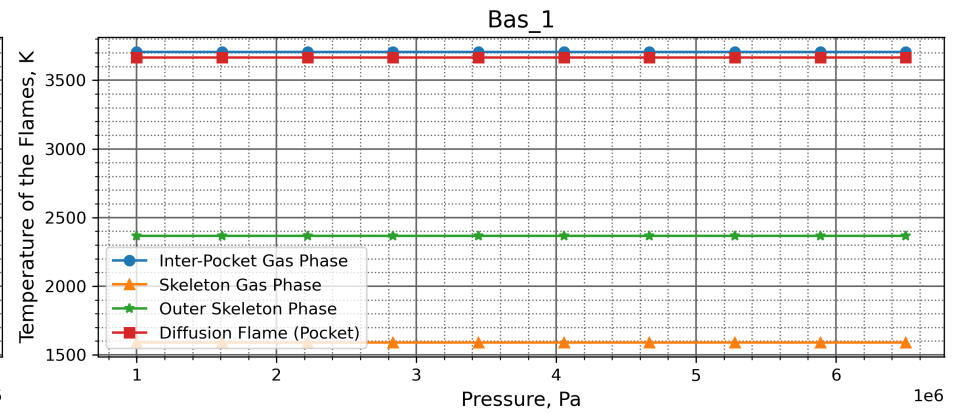
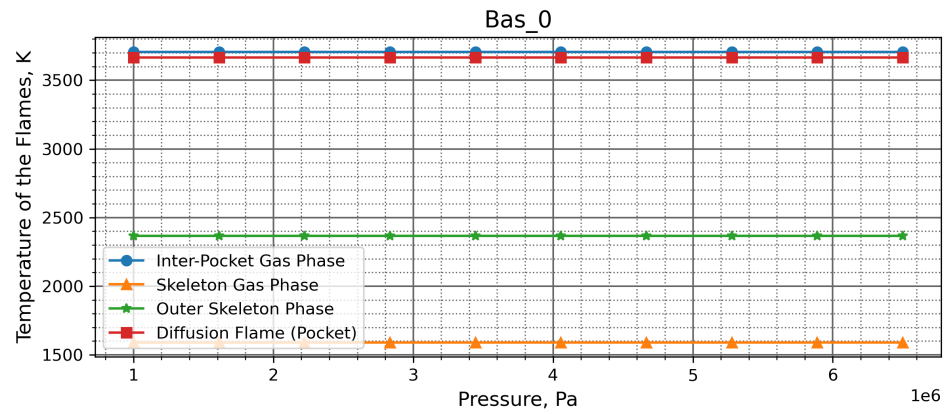
Average Molar Mass, kg/mol



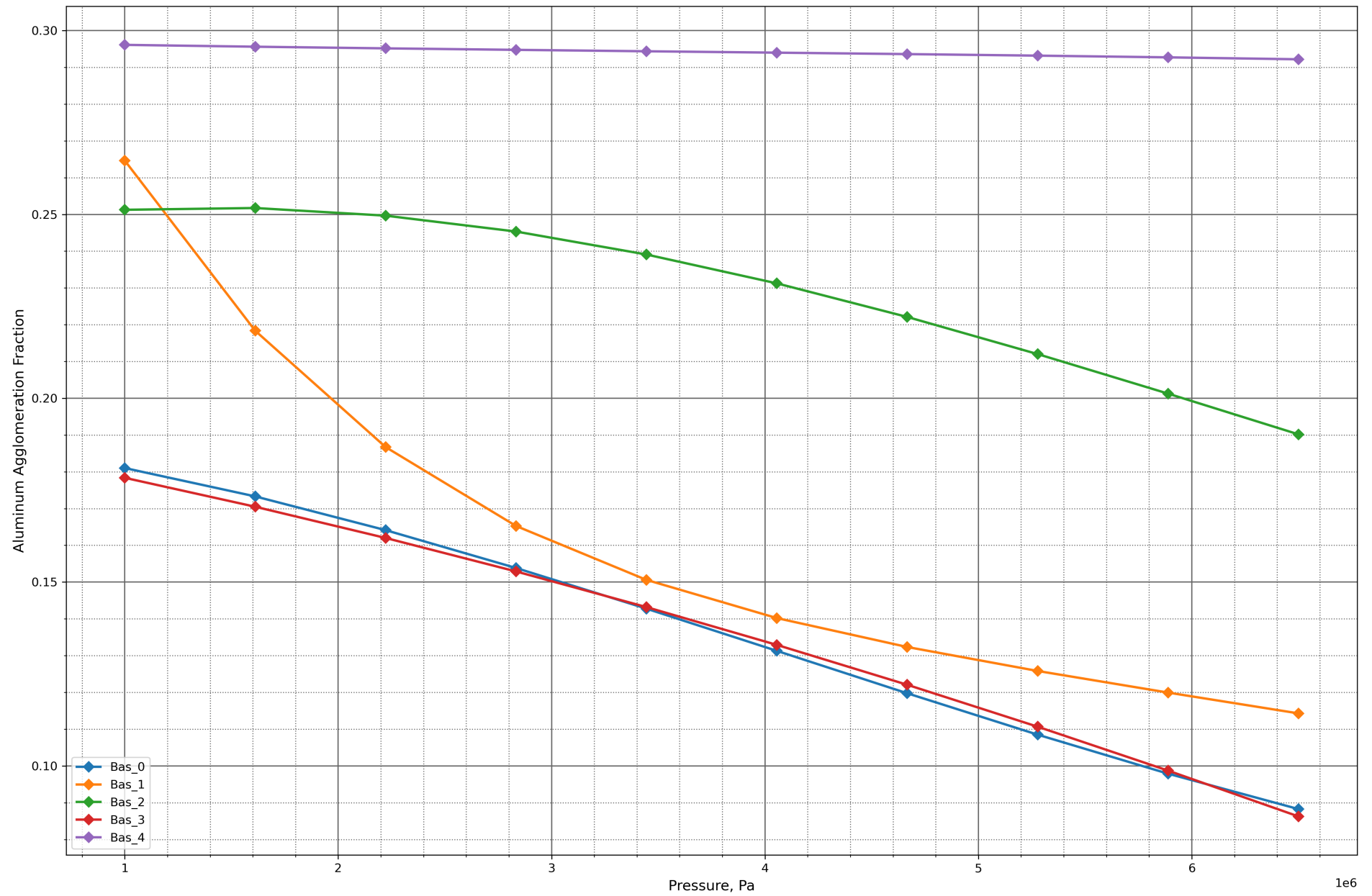
Constant Volume Heat Capacity (c_v), J/(kg·K)



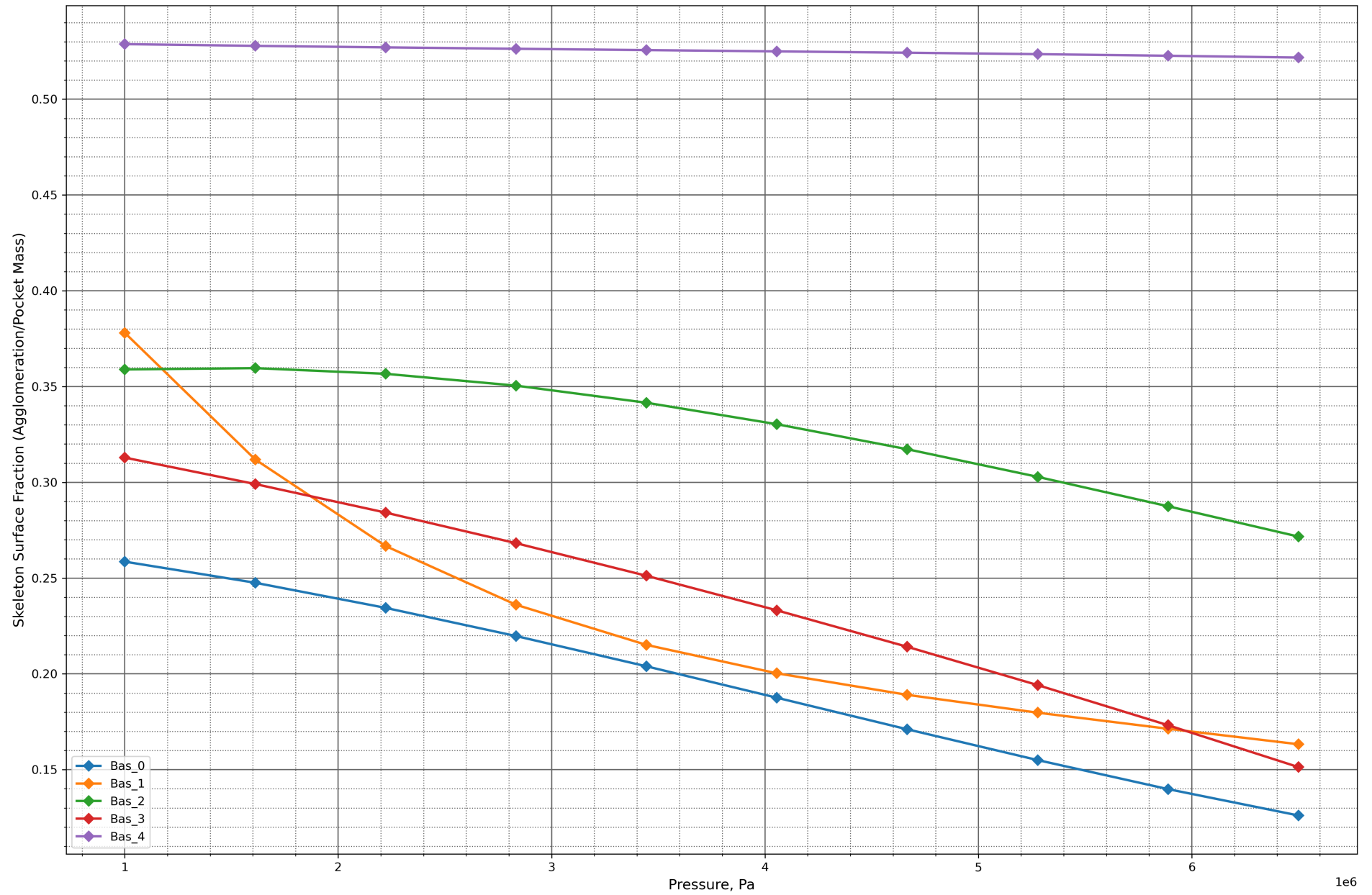
Temperature of the Flames, K



Aluminum Agglomeration Fraction



Skeleton Surface Fraction (Agglomeration/Pocket Mass)



Performance Report

System Information

Operating System: Unix 6.8.0.124 (Ubuntu 24.04.4 LTS)
Processor: Intel(R) Xeon(R) CPU E5-2697 v4 @ 2.30GHz (Base Clock Frequency 2.29 GHz)
Memory: 15.6 GB (64-bit), Unknown
Machine Name: vscode
Runtime: .NET 10.0.9 (.NET 10.0.9)
Architecture: X64 / X64

Total Execution Time

Total execution time: 05:53:42.296

Garbage Collection Information

Total Memory: 7125704 bytes
GC Pause Time Percentage: 0.12%
Generation 0 Collections: 22370
Generation 1 Collections: 1892
Generation 2 Collections: 211

Detailed Performance Metrics

Method: Telemetry/Proc/0/OptimizationProblemExecutionPerformance

Call count: 11103763
Maximum execution time: 7.8408 ms
Minimum execution time: 0.0001 ms
Mean execution time: 0.9550821028421154 ms
Execution time standard deviation: 0.26377021768486913 ms

Method: Telemetry/Proc/0/OptimizationProblemExecutionPerformance

Call count: 10391366
Maximum execution time: 7.169 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.35658610851547523 ms
Execution time standard deviation: 0.07918381428554402 ms

Method: Telemetry/Proc/0/OptimizationProblemExecutionPerformance

Call count: 10151914
Maximum execution time: 4.5092 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.33402305991758785 ms
Execution time standard deviation: 0.11228889371502405 ms

Method: Telemetry/Proc/1/OptimizationProblemExecutionPerformance

Call count: 11003292
Maximum execution time: 7.9869 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.9748519275595264 ms
Execution time standard deviation: 0.2611381595311732 ms

Method: Telemetry/Proc/1/OptimizationProblemExecutionPerformance

Call count: 10273606
Maximum execution time: 12.9329 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.3639913369073941 ms
Execution time standard deviation: 0.07622945642768025 ms

Method: Telemetry/Proc/1/OptimizationProblemExecutionPerformance

Call count: 10028606
Maximum execution time: 4.4446 ms

Minimum execution time: 0.0002 ms
Mean execution time: 0.3400783043824712 ms
Execution time standard deviation: 0.11285945418478657 ms

Method: Telemetry/Proc/2/OptimizationProblemExecutionPerformance
Call count: 11003292
Maximum execution time: 13.1997 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.9387585182416612 ms
Execution time standard deviation: 0.2630567182936548 ms

Method: Telemetry/Proc/2/OptimizationProblemExecutionPerformance
Call count: 10313942
Maximum execution time: 7.2501 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.3499431388794318 ms
Execution time standard deviation: 0.07967941826859291 ms

Method: Telemetry/Proc/2/OptimizationProblemExecutionPerformance
Call count: 10082230
Maximum execution time: 4.3788 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.3292130978365629 ms
Execution time standard deviation: 0.11244125352563615 ms

Method: Telemetry/Proc/3/OptimizationProblemExecutionPerformance
Call count: 11003292
Maximum execution time: 13.6103 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.9313370384609365 ms
Execution time standard deviation: 0.2657313994277633 ms

Method: Telemetry/Proc/3/OptimizationProblemExecutionPerformance
Call count: 10298456
Maximum execution time: 6.6362 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.3467342044768279 ms
Execution time standard deviation: 0.08087136338684728 ms

Method: Telemetry/Proc/3/OptimizationProblemExecutionPerformance
Call count: 10057581
Maximum execution time: 5.0744 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.3254000368378752 ms
Execution time standard deviation: 0.11389614734617642 ms

Method: Telemetry/Proc/4/OptimizationProblemExecutionPerformance
Call count: 11003292
Maximum execution time: 9.8023 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.9278880342809798 ms
Execution time standard deviation: 0.27233453308909 ms

Method: Telemetry/Proc/4/OptimizationProblemExecutionPerformance
Call count: 10250760
Maximum execution time: 7.8789 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.34718806375333683 ms
Execution time standard deviation: 0.08278670310060746 ms

Method: Telemetry/Proc/4/OptimizationProblemExecutionPerformance

Call count: 10000320

Maximum execution time: 4.95 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.3245566006687558 ms

Execution time standard deviation: 0.11488943806297575 ms

Method: Telemetry/Proc/5/OptimizationProblemExecutionPerformance

Call count: 11003292

Maximum execution time: 7.5557 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.9183708097540918 ms

Execution time standard deviation: 0.27165639252879775 ms

Method: Telemetry/Proc/5/OptimizationProblemExecutionPerformance

Call count: 10294476

Maximum execution time: 4.6491 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.34279225485596143 ms

Execution time standard deviation: 0.08240633718440536 ms

Method: Telemetry/Proc/5/OptimizationProblemExecutionPerformance

Call count: 10050239

Maximum execution time: 4.4973 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.3221630343219486 ms

Execution time standard deviation: 0.1126805739318281 ms

Method: Telemetry/Proc/6/OptimizationProblemExecutionPerformance

Call count: 11003292

Maximum execution time: 7.7158 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.9689582462867029 ms

Execution time standard deviation: 0.2628148213174156 ms

Method: Telemetry/Proc/6/OptimizationProblemExecutionPerformance

Call count: 10268574

Maximum execution time: 4.6019 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.3634125847951548 ms

Execution time standard deviation: 0.07748580732894501 ms

Method: Telemetry/Proc/6/OptimizationProblemExecutionPerformance

Call count: 10012425

Maximum execution time: 4.7672 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.3430091893821585 ms

Execution time standard deviation: 0.11461749765940506 ms

Method: Telemetry/Proc/7/OptimizationProblemExecutionPerformance

Call count: 11003292

Maximum execution time: 5.6765 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.910429504379308 ms

Execution time standard deviation: 0.2674927554113324 ms

Method: Telemetry/Proc/7/OptimizationProblemExecutionPerformance

Call count: 10259504

Maximum execution time: 7.7469 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.33953327355782975 ms
Execution time standard deviation: 0.08181059243018433 ms

Method: Telemetry/Proc/7/OptimizationProblemExecutionPerformance

Call count: 9974609

Maximum execution time: 4.5008 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.3186367179505504 ms
Execution time standard deviation: 0.11262587283741901 ms

Method: Telemetry/Proc/8/OptimizationProblemExecutionPerformance

Call count: 11003292

Maximum execution time: 7.2033 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.9395102438615593 ms
Execution time standard deviation: 0.26523740833944587 ms

Method: Telemetry/Proc/8/OptimizationProblemExecutionPerformance

Call count: 10281072

Maximum execution time: 6.803 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.3505965172600881 ms
Execution time standard deviation: 0.08022265113917355 ms

Method: Telemetry/Proc/8/OptimizationProblemExecutionPerformance

Call count: 10037076

Maximum execution time: 6.1422 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.32967114385694735 ms
Execution time standard deviation: 0.1136187567105907 ms

Method: Telemetry/Proc/9/OptimizationProblemExecutionPerformance

Call count: 11003292

Maximum execution time: 7.3975 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.9468931781324663 ms
Execution time standard deviation: 0.26653912471537 ms

Method: Telemetry/Proc/9/OptimizationProblemExecutionPerformance

Call count: 10324908

Maximum execution time: 7.1606 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.3537437482445634 ms
Execution time standard deviation: 0.08049580408561871 ms

Method: Telemetry/Proc/9/OptimizationProblemExecutionPerformance

Call count: 10066695

Maximum execution time: 4.8748 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.33133810158154664 ms
Execution time standard deviation: 0.11355076468485287 ms

Method: Telemetry/Proc/10/OptimizationProblemExecutionPerformance

Call count: 11003292

Maximum execution time: 11.9633 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.9578496102711972 ms

Execution time standard deviation: 0.2644041968723326 ms

Method: Telemetry/Proc/10/OptimizationProblemExecutionPerformance

Call count: 10270748

Maximum execution time: 6.7964 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.35954345338826826 ms

Execution time standard deviation: 0.07946682830096173 ms

Method: Telemetry/Proc/10/OptimizationProblemExecutionPerformance

Call count: 10021044

Maximum execution time: 4.2866 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.33444600045665157 ms

Execution time standard deviation: 0.11347434622506275 ms

Method: Telemetry/Proc/11/OptimizationProblemExecutionPerformance

Call count: 11003292

Maximum execution time: 14.5339 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.9108175151400315 ms

Execution time standard deviation: 0.26300174125620956 ms

Method: Telemetry/Proc/11/OptimizationProblemExecutionPerformance

Call count: 10300384

Maximum execution time: 6.4941 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.3394268966865654 ms

Execution time standard deviation: 0.08086797764636133 ms

Method: Telemetry/Proc/11/OptimizationProblemExecutionPerformance

Call count: 10062013

Maximum execution time: 12.728 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.3168502094660077 ms

Execution time standard deviation: 0.11184914458589787 ms

Method: Telemetry/Proc/12/OptimizationProblemExecutionPerformance

Call count: 11003292

Maximum execution time: 8.0582 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.9479718334656873 ms

Execution time standard deviation: 0.2632292905506158 ms

Method: Telemetry/Proc/12/OptimizationProblemExecutionPerformance

Call count: 10292339

Maximum execution time: 5.5694 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.3526556793941435 ms

Execution time standard deviation: 0.07885893119342001 ms

Method: Telemetry/Proc/12/OptimizationProblemExecutionPerformance

Call count: 10057114

Maximum execution time: 4.2829 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.3300803571183355 ms

Execution time standard deviation: 0.11069716790430942 ms

Method: Telemetry/Proc/13/OptimizationProblemExecutionPerformance

Call count: 11003292
Maximum execution time: 8.1414 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.9703151034072753 ms
Execution time standard deviation: 0.2638410691601877 ms

Method: Telemetry/Proc/13/OptimizationProblemExecutionPerformance

Call count: 10278973
Maximum execution time: 11.4787 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.36318092191701856 ms
Execution time standard deviation: 0.07647136466262064 ms

Method: Telemetry/Proc/13/OptimizationProblemExecutionPerformance

Call count: 10043614
Maximum execution time: 6.2 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.3423124000882115 ms
Execution time standard deviation: 0.11189651016224109 ms

Method: Telemetry/Proc/14/OptimizationProblemExecutionPerformance

Call count: 11003292
Maximum execution time: 20.4863 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.9601165439032532 ms
Execution time standard deviation: 0.26553121635921395 ms

Method: Telemetry/Proc/14/OptimizationProblemExecutionPerformance

Call count: 10303153
Maximum execution time: 4.2388 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.3579337800671144 ms
Execution time standard deviation: 0.07900126939761841 ms

Method: Telemetry/Proc/14/OptimizationProblemExecutionPerformance

Call count: 10062623
Maximum execution time: 4.8465 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.3341676820248205 ms
Execution time standard deviation: 0.1137094222693365 ms

Method: Telemetry/Proc/15/OptimizationProblemExecutionPerformance

Call count: 11003292
Maximum execution time: 8.2146 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.9387451601938953 ms
Execution time standard deviation: 0.2705649623022228 ms

Method: Telemetry/Proc/15/OptimizationProblemExecutionPerformance

Call count: 10278876
Maximum execution time: 7.4633 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.3503687004590811 ms
Execution time standard deviation: 0.08183352318515974 ms

Method: Telemetry/Proc/15/OptimizationProblemExecutionPerformance

Call count: 10021452
Maximum execution time: 4.9474 ms
Minimum execution time: 0.0002 ms

Mean execution time: 0.32861819035810225 ms
Execution time standard deviation: 0.11401677392295773 ms

Method: Telemetry/Proc/16/OptimizationProblemExecutionPerformance
Call count: 11003292
Maximum execution time: 7.9903 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.9398587504722036 ms
Execution time standard deviation: 0.26024515680117744 ms

Method: Telemetry/Proc/16/OptimizationProblemExecutionPerformance
Call count: 10340322
Maximum execution time: 4.6051 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.3485151240357756 ms
Execution time standard deviation: 0.07925850893279883 ms

Method: Telemetry/Proc/16/OptimizationProblemExecutionPerformance
Call count: 10113696
Maximum execution time: 6.7808 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.3241583628774481 ms
Execution time standard deviation: 0.11108079037964236 ms

Method: Telemetry/Proc/17/OptimizationProblemExecutionPerformance
Call count: 11003292
Maximum execution time: 8.3274 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.9282470807008969 ms
Execution time standard deviation: 0.2690227480174758 ms

Method: Telemetry/Proc/17/OptimizationProblemExecutionPerformance
Call count: 10280667
Maximum execution time: 4.3151 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.3476548731419974 ms
Execution time standard deviation: 0.08038492612263115 ms

Method: Telemetry/Proc/17/OptimizationProblemExecutionPerformance
Call count: 10032012
Maximum execution time: 4.7183 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.3285319159805276 ms
Execution time standard deviation: 0.11266761947163029 ms

Method: Telemetry/Proc/18/OptimizationProblemExecutionPerformance
Call count: 11003292
Maximum execution time: 19.0994 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.9548080183457578 ms
Execution time standard deviation: 0.2615008161629259 ms

Method: Telemetry/Proc/18/OptimizationProblemExecutionPerformance
Call count: 10302871
Maximum execution time: 4.4226 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.3558295870831176 ms
Execution time standard deviation: 0.07745562673221097 ms

Method: Telemetry/Proc/18/OptimizationProblemExecutionPerformance
Call count: 10062847
Maximum execution time: 4.598 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.33309922349013027 ms
Execution time standard deviation: 0.11035994849792956 ms

Method: Telemetry/Proc/19/OptimizationProblemExecutionPerformance
Call count: 11003292
Maximum execution time: 7.9108 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.9304644685699665 ms
Execution time standard deviation: 0.26571710551382 ms

Method: Telemetry/Proc/19/OptimizationProblemExecutionPerformance
Call count: 10260865
Maximum execution time: 4.66 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.34804698982985566 ms
Execution time standard deviation: 0.07937402807049956 ms

Method: Telemetry/Proc/19/OptimizationProblemExecutionPerformance
Call count: 10014103
Maximum execution time: 6.6937 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.32532814919121933 ms
Execution time standard deviation: 0.11324241482869693 ms

Method: Telemetry/Proc/20/OptimizationProblemExecutionPerformance
Call count: 11003292
Maximum execution time: 6.9127 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.9409786900320243 ms
Execution time standard deviation: 0.26083651982596684 ms

Method: Telemetry/Proc/20/OptimizationProblemExecutionPerformance
Call count: 10310339
Maximum execution time: 7.1119 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.3505560008841302 ms
Execution time standard deviation: 0.07910695202207643 ms

Method: Telemetry/Proc/20/OptimizationProblemExecutionPerformance
Call count: 10070106
Maximum execution time: 7.3808 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.3279140215703694 ms
Execution time standard deviation: 0.11249582168331235 ms

Method: Telemetry/Proc/21/OptimizationProblemExecutionPerformance
Call count: 11003292
Maximum execution time: 11.7991 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.9253506608295305 ms
Execution time standard deviation: 0.2692080197213122 ms

Method: Telemetry/Proc/21/OptimizationProblemExecutionPerformance
Call count: 10248519
Maximum execution time: 7.8026 ms

Minimum execution time: 0.0002 ms
Mean execution time: 0.34614759790173166 ms
Execution time standard deviation: 0.0823323404053972 ms

Method: Telemetry/Proc/21/OptimizationProblemExecutionPerformance
Call count: 10001892
Maximum execution time: 17.9898 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.3241869980200222 ms
Execution time standard deviation: 0.11350338139331279 ms

Method: Telemetry/Proc/22/OptimizationProblemExecutionPerformance
Call count: 11003292
Maximum execution time: 12.025 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.963062036788647 ms
Execution time standard deviation: 0.26652416481488655 ms

Method: Telemetry/Proc/22/OptimizationProblemExecutionPerformance
Call count: 10301429
Maximum execution time: 6.1387 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.35903134009858745 ms
Execution time standard deviation: 0.07945070314872833 ms

Method: Telemetry/Proc/22/OptimizationProblemExecutionPerformance
Call count: 10054554
Maximum execution time: 4.5649 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.33691164186892025 ms
Execution time standard deviation: 0.11355976858848857 ms

Method: Telemetry/Proc/23/OptimizationProblemExecutionPerformance
Call count: 11003292
Maximum execution time: 14.3837 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.9418582435965246 ms
Execution time standard deviation: 0.263764841209533 ms

Method: Telemetry/Proc/23/OptimizationProblemExecutionPerformance
Call count: 10298947
Maximum execution time: 4.6303 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.35061902345937324 ms
Execution time standard deviation: 0.07896178458395867 ms

Method: Telemetry/Proc/23/OptimizationProblemExecutionPerformance
Call count: 10063864
Maximum execution time: 4.5938 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.32917572465212563 ms
Execution time standard deviation: 0.11185381322608129 ms

Method: Telemetry/Proc/24/OptimizationProblemExecutionPerformance
Call count: 11003292
Maximum execution time: 7.0523 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.9788671476318469 ms
Execution time standard deviation: 0.2601809074717658 ms

Method: Telemetry/Proc/24/OptimizationProblemExecutionPerformance

Call count: 10279117

Maximum execution time: 5.8143 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.3650188799291173 ms

Execution time standard deviation: 0.07536855629903486 ms

Method: Telemetry/Proc/24/OptimizationProblemExecutionPerformance

Call count: 10033891

Maximum execution time: 4.0242 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.3396741459320386 ms

Execution time standard deviation: 0.11166257881471474 ms

Method: Telemetry/Proc/25/OptimizationProblemExecutionPerformance

Call count: 11003292

Maximum execution time: 6.9924 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.9209605241868433 ms

Execution time standard deviation: 0.26434430409916954 ms

Method: Telemetry/Proc/25/OptimizationProblemExecutionPerformance

Call count: 10274762

Maximum execution time: 4.4775 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.3425108821012638 ms

Execution time standard deviation: 0.08017821527598747 ms

Method: Telemetry/Proc/25/OptimizationProblemExecutionPerformance

Call count: 10026296

Maximum execution time: 4.2704 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.32013251540748977 ms

Execution time standard deviation: 0.11103829577907597 ms